

A MODEL OF OPTIMAL SCIENCE FUNDING: TARGETING THE CAPABILITY-RESOURCE GAP

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ABSTRACT. The grant funder’s challenge is to maximize the impact of a fixed budget under uncertainty about the researchers who might receive it. We develop a Bayesian model of this challenge: researchers differ in their capabilities and in their resources; research output requires both, and funding compounds capability across rounds. We show that the output-maximizing allocation of a round’s budget funds the capability-resource gap: grants increase with capability and decrease with existing resources. Track records do not pin down the gap, because research output confounds the two; what a funder needs is a signal of capability less confounded by resources than the track record, and peer review of a proposal can provide one. We find that the value of that signal increases with how unequally the gaps are distributed, which capability inequality mainly drives, and with the signal’s informativeness, and that under heavy-tailed capability distributions even a moderately informative signal captures most of that value. Uniform seed grants and lotteries, which forgo targeting, lose little research output exactly where review is worth little, and where informative review makes a partial lottery costly, most of what the lottery loses comes from discarding the review ranking rather than from chance. Two features of a field, the funder’s budget relative to researchers’ resources and the inequality of the gaps, locate a funder on a map of these results.

Significance. Research funders divide fixed budgets among researchers whose abilities and needs they cannot observe; whether to do so by peer review, by lottery, or by spreading funds is still debated. We model funding as a sequential decision under uncertainty; the output-maximizing allocation funds the gap between each researcher’s capability and their resources, a gap that track records alone do not pin down. What peer review is worth, and how seed grants and lotteries perform, turn on three features a funder can estimate: its budget relative to researchers’ resources, how unequal capability is in its field, and how informative its review is. A small foundation in an unequal field should target; a large agency in an even field can spread funds.

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The United States spent close to a trillion dollars on research and development in 2024, roughly a fifth of it federal (National Center for Science and Engineering Statistics 2026; National Science Board 2026), and how best to allocate that money is still debated. Whether funding should be concentrated among a few researchers or spread across many is a long-running dispute (Fortin and Currie 2013; Mongeon et al. 2016; Aagaard et al. 2020; Lauer and Roychowdhury 2021). How well peer review predicts research outcomes is contested: among funded NIH grants, percentile scores predicted subsequent productivity barely better than chance (Fang et al. 2016), reviewers scoring the same application agree only weakly (Pier et al. 2018), and studies with other designs find modest but real predictive power (Li and Agha 2015; Park et al. 2015; Lindner and Nakamura 2015). Several funders now allocate part of their budgets by lottery, from a tie-break among top-ranked proposals to a draw among all applicants above a threshold (Liu et al. 2020; Heyard et al. 2022; Feliciani et al. 2024). A randomized trial of winning a grant found no clear effect on the recipients’ publications over four years (Barnett et al. 2024), and a quasi-experimental study of NIH grants found a small one (Jacob and Lefgren 2011), which raises the more basic question of when an additional grant adds value at all.

These disputes concern one problem: what a funder with a fixed budget and imperfect knowledge of researchers should do. Each empirical study measures one input to that decision; the answer depends on what the funder is uncertain about, what its information is worth, and how the budget changes both. We treat the funder’s problem as a sequential decision under uncertainty and analyze it with a Bayesian approach. Researchers differ in capability, what they know and can do, and in resources, what they have to work with. Research output requires both and is limited by the scarcer of the two. The funder observes research output and the noisy judgments of reviewers, but neither attribute directly. A grant adds resources in the round it is given and, because doing research builds expertise, raises capability in later rounds. Each round, the funder observes the research output, updates its beliefs, and allocates again.

Three results follow. First, the allocation of a round’s budget that maximizes expected research output funds the gap between each researcher’s capability and their resources: grants increase with capability and decrease with existing resources, and the budget sets how many researchers are funded at all. Two heuristics a funder might use, funding the track record and funding the under-resourced, each track one of the two quantities, and each can perform worse than dividing the budget uniformly across all researchers. Second, track records do not pin down the gap, because research output confounds capability with resources; what the

funder needs is a signal of capability less confounded by resources than the track record, and peer review of a proposal can provide one. The value of that signal, and how seed grants and lotteries perform relative to targeted funding, depend on two features of a field: how much of researchers' resources the funder's budget can supply, and how unequally the capability-resource gaps are distributed, which in every setting we considered is mostly a matter of how unequal capability is. Where the budget is tight and capability heavy-tailed, targeting is worth a great deal, even moderately noisy review captures most of that value, and lotteries give up much of it; where the budget is ample and capability evenly spread, little research output is at stake and spreading funds widely loses almost nothing. Third, the two features can be estimated from what funders already know, and we place representative funding programs on the budget axis of the map they define.

Two assumptions bound these results: grants are used up in the round they are given, with whatever lasts from them entering as capability, and capability and resources are complements. The objective is aggregate expected research output; diversity of directions, risk, and fairness are outside it, and "optimal" throughout means output-maximizing under that objective. The model is deliberately simple. It shows that the disputes turn on two features of a field, and which way each dispute goes in each kind of field.

RESULTS

The optimal allocation funds the capability-resource gap. Consider first a funder that knows every researcher's capability K_i and resources R_i and divides one round's budget B so as to maximize that round's expected research output, which for researcher i is $AK_i R_i / (K_i + R_i)$, where A sets the units of research output and R_i is the sum of baseline resources R_{i0} and the grant g_i (Materials and Methods). The marginal value of a dollar to researcher i is $AK_i^2 / (K_i + R_{i0} + g_i)^2$: it increases with their capability and decreases with the resources they already hold, grants included. An allocation is optimal exactly when no dollar can be moved to a researcher who would produce more with it, and that condition gives the gap rule (SI Text 1, Proposition 1):

$$g_i^* = \max(c K_i - R_{i0}, 0),$$

where $c > 0$ is the value at which the grants sum to the budget. Each researcher's target is c times their capability; the rule fills each positive gap between target and resources and grants nothing to the rest. The rule holds for any production function (the rule that maps capability and resources to expected research output) with constant returns to scale and diminishing returns to resources, $F(K, R) = A K h(R/K)$ with h increasing and strictly concave, because equal marginal value

among the funded then means a common ratio of resources to capability (SI Text 1, Proposition 1); the harmonic mean is the instance we simulate. The condition $R_{i0} < cK_i$ draws a frontier through the population, separating the funded from the unfunded (Fig. 1). A larger budget raises c , moves the frontier outward, and funds more researchers with larger grants, until funding extends across nearly the whole population.

The gap rule explains why each heuristic fails. Funding the track record fails because research output reflects resources as well as capability: a productive, well-resourced researcher has a small or negative gap, and an additional dollar adds little to their research output. Funding the under-resourced fails for the opposite reason: scarcity is no evidence of capability, and dollars given to researchers of low capability add little research output however scarce their resources. Each heuristic can produce less research output than dividing the budget equally (SI Text 1, Examples 1 and 2; Fig. S1).

How much targeting matters depends on the budget. Expected research output never exceeds AK_i , so when the budget is large enough to bring every researcher’s research output near that bound, the optimal allocation and equal division differ little. The value of targeting, the expected research output the optimal allocation adds over equal division, vanishes as the budget grows whenever capability bounds research output, as it does under the harmonic mean (SI Text 1, Corollary 2), and in every one of several hundred simulated populations it also fell with the budget throughout, as a share of uniform funding’s own gain (Fig. S2). This dependence on the budget recurs in every result below.

Proposition 1 concerns one round. Over several rounds a grant also raises the recipient’s future capability, and the allocation that maximizes research output over the whole horizon need not take the gap form. Numerically it nearly does: over two rounds with complete information, the two-round optimum, free to choose its recipients and how much of the budget to spend in each round, exceeds the gap rule applied round by round with an even split by about one percent of the research output that funding adds or less, at every capability tail, compounding rate, and budget in our parameter sweep, and the two allocations’ first-round grants are nearly identical (SI Text 1, Table S4). We therefore use the round-by-round gap rule as the complete-information benchmark, and “optimal allocation” below refers to it.

What the funder can learn: track records and peer review. A real funder observes research output, not capability or resources. Expected research output is one number produced by the two jointly, so a researcher of high capability and modest resources and a researcher of modest capability and abundant resources

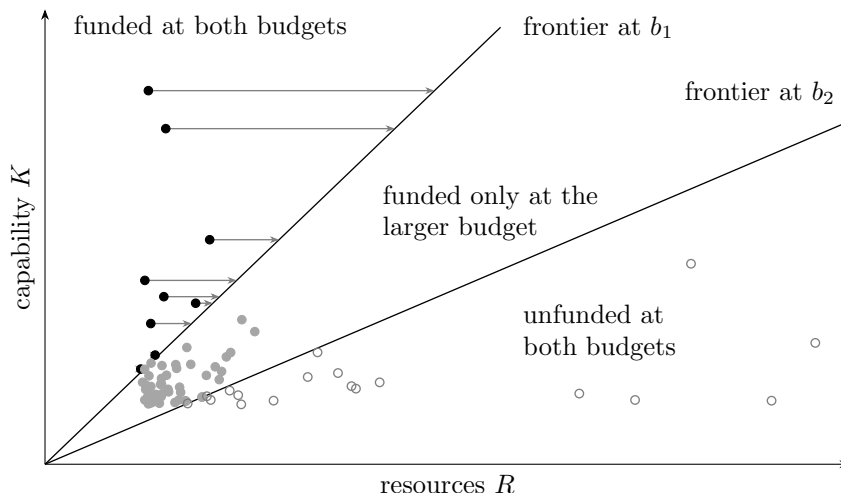


FIGURE 1. The gap rule in one population at two budgets. Each point is a researcher, placed by resources and capability. The two lines are the funding frontiers $R = cK$ at a smaller budget b_1 and a larger budget b_2 ; a larger budget raises c and flattens the frontier. Researchers to the left of a frontier are funded at that budget: black points at both budgets, gray points only at the larger budget, open points at neither. Each arrow is a grant at the smaller budget, which moves its researcher exactly to that budget's frontier; at the larger budget every black and gray researcher receives a grant that moves it to the outer frontier (arrows not drawn).

can produce identical records, and the gap rule treats these two most differently: the first may have the largest gap in the population, the second none. Expected research output gives only a lower bound on capability, $K_i > \lambda_i/A$, and realized research output, a Poisson draw around λ_i , bounds it less. The problem is worst where the gap is largest. When resources are small relative to capability, expected research output is approximately AR_i , proportional to resources and nearly independent of capability (Fig. 2A), so on small grants research output carries little information about capability; only grants comparable to capability itself make records informative about it.

A Bayesian funder without review, which holds the track record and the noisy resource signal but no review score (Materials and Methods), therefore allocates far from the gap rule, and further rounds of records improve its allocation only slowly: its expected research output falls short of the complete-information optimum by nearly half of what that optimum adds in the first round and still by about a fifth in round twenty, while a funder that also holds a single review score for each researcher, observed once at the start, starts under a tenth short and stays there

(Fig. 2*B*). Capabilities compound while resources do not, so each researcher’s research output approaches the level their resources sustain, and research output at that level carries less and less information about capability.

What the funder lacks is a signal of capability less confounded by resources than the track record. Peer review of a proposal can provide one. We model it as each researcher’s capability plus Gaussian noise, observed once (Materials and Methods); a score that partly reflects resources is a robustness check below. The value of review is the additional expected research output of a funder allocating with the signal over the same funder without it, the value of the information the score carries (Howard 1966), reported as the share it recovers of the no-review shortfall, the research output that complete information would add over the funder without review. That value depends on the field (Fig. 2*C*). Review is worth most where capability is heavy-tailed and the budget is tight, and under heavy-tailed capability even a signal of moderate informativeness captures most of its value: at the default signal noise, review retains most of its maximum value where capability is heavy-tailed (tail parameter $\alpha_K = 1.3$) and under half where capability is spread more evenly ($\alpha_K = 3.5$), with the intermediate distribution ($\alpha_K = 2$) between them; at three times that noise, about half and almost none (Fig. 2*C*). The three distributions have the same mean capability, so the tail parameter varies how dispersed and how heavy-tailed capability is, not its level. The reason is twofold. Where capability is heavy-tailed, most of the research output a funder can add comes from getting resources to the few researchers of unusually high capability, and precisely those researchers are the easiest to identify: they stand far apart from everyone else, so even a noisy signal separates them. Where capability is spread more evenly, no researcher matters much more than another, so review of any informativeness adds little; and where the budget is ample, information about whom to fund has little left to add.

This ordering of review’s value by field holds across production functions from Cobb-Douglas to Leontief (SI Text 4), at equal rank correlation between score and capability rather than equal noise, for applicant pools of 25 to 200, for heavier and lighter resource tails, when the review score partly reflects resources, when review is redrawn every round, and across resource signals from nearly exact to nearly uninformative (SI Text 5). Multiplicative review noise, a constant relative error, narrows the heavy-tailed field’s advantage without removing it, because a constant relative error is a large absolute error for exactly the researchers who matter most there. What review supplies in every variant is information about capability that research output cannot supply on its own; peer review is one source, and the results apply to any other, such as a pilot grant large enough to make

research output informative. The association between capability and resources changes how much any of this is worth: where the capable already hold the resources, the gaps are small and even, and both targeting and review are worth less. One condition qualifies the value of review: the funder must weight review at its true informativeness. A funder that overtrusts review, treating a noisy signal as reliable, can do worse than one that ignores review altogether, and overtrusting review loses more research output than undertrusting it (Fig. S3).

Spreading funds: seed grants, lotteries, and concentration. Two alternatives to targeting recur in discussions of funding policy. Uniform seed grants give every researcher the same amount, so that a researcher whom review misjudges still receives something; in practice a funder gives out part of its budget as seed grants and targets the rest. Lotteries give grants to applicants chosen at random, usually from among those who pass a review threshold; their advocates argue that review cannot reliably rank the applicants it deems fundable (Pier et al. 2018; Shaw 2023), that review is costly (Herbert et al. 2013; Gross and Bergstrom 2019), and that review is biased (Ginther et al. 2011; Wittelman et al. 2019; Boudreau et al. 2016; Fang and Casadevall 2016; Avin 2019a; Roumbanis 2019); a parallel case on the same grounds has been made for abolishing pre-publication peer review of articles (Heesen and Bright 2021). Both give up part of the value of targeting on purpose, and that value depends on the field: for a funder allocating with records and review, targeting adds more research output than uniform funding’s entire gain over no funding in a heavy-tailed field with reliable review and a tight budget, and about a quarter of that gain in the intermediate field (capability tail parameter 2, between the heavy-tailed and evenly spread fields) at the largest budget in our parameter sweep (SI Text 3). What the alternatives save, in review costs and proposal effort, lies outside our model (Gross and Bergstrom 2019).

Consider seed grants first. The research output lost increases faster than the fraction of the budget given out as seed grants: half the budget as seed grants loses more than twice what a quarter loses (Fig. 3A). The size of the loss is set by what targeting is worth in the field: a review-informed funder that gives out three quarters of its budget as seed grants forfeits about a tenth of the research output its funding adds in a heavy-tailed field with reliable review, more at the smallest budget in our parameter sweep, and a few percent in the intermediate field.

Now consider lotteries. The lotteries funders run are partial: review sets a fundable pool, and chance picks the grantees within it. Such a lottery gives up the review ranking within the pool and the certainty of who is funded; we ask what each costs. We evaluate five schemes in a single round (Fig. 3B): equal division among

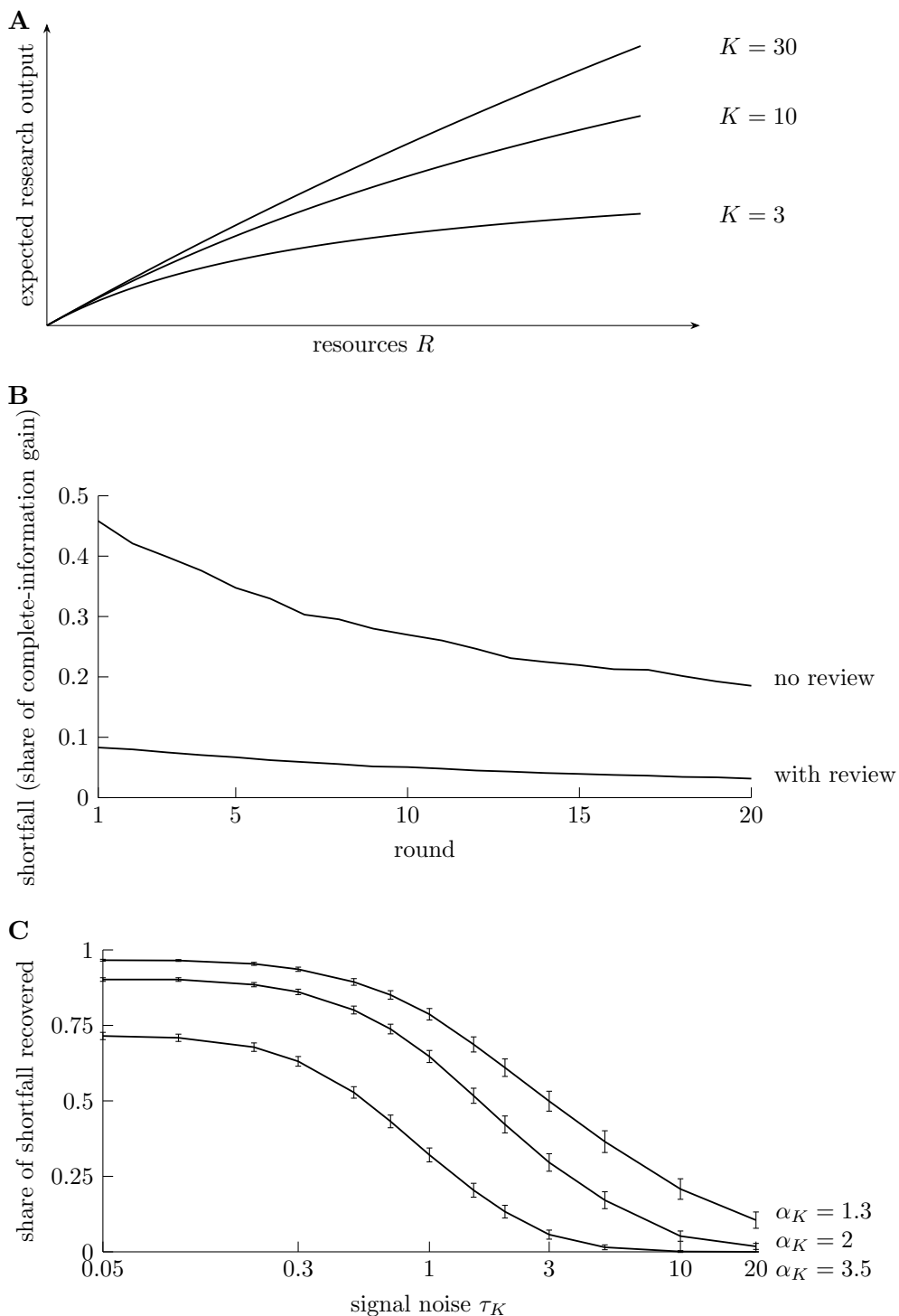


FIGURE 2. What the funder can learn. (A) On small grants, research output carries little information about capability: expected research output against resources for researchers of capability $K = 3, 10$, and 30 ; the curves coincide where resources are small relative to capability. (B) Records do not catch up: the shortfall of each funder's expected research output below the complete-information optimum, as a share of what complete-information funding adds, round by round in the intermediate field; the no-review funder's shortfall falls from 46% in round one to 19% by round twenty, while the funder with a single review score observed at the start is at 8% in round one and 3% by round twenty. (C)

the fifth of researchers with the highest review scores (selection by review); equal division among the top two fifths; the partial lottery that some funders run, which funds half of the top two fifths at random with full-size grants; uniform funding; and a lottery over all researchers with full-size grants. In a heavy-tailed field with a tight budget, selection by review gains about three quarters of what the optimal allocation gains when review is reliable and about half when review is very noisy. The partial lottery gains less, and most of the difference comes from ignoring the review ranking within the top two fifths: at the default noise, the ignored ranking accounts for most of the partial lottery’s shortfall below selection by review and the randomness for a small part. The lottery over all researchers gains least, less than uniform funding. In an evenly spread field with a large budget, the ordering reverses: uniform funding gains more than four fifths of what the optimal allocation gains, and every scheme that concentrates grants gains less. A lottery among applicants above a review threshold therefore loses little research output only where the review ranking it ignores is worth little: a few hundredths of the optimal allocation’s gain in an evenly spread field or when review is very noisy, about a sixth of that gain in a heavy-tailed field with reliable review. In an evenly spread field, the better alternative to review is small grants to many researchers.

The randomness itself costs research output for a general reason. A lottery gives each applicant in its pool a chance at a full grant instead of a certain smaller one of the same expected size, and because expected research output increases with resources at a diminishing rate, the certain grant produces more. Dividing a pool’s money equally therefore produces more expected research output than any lottery over the same pool, provided that grants are divisible and that research output rises with resources at a diminishing rate at every grant size (SI Text 2, Proposition 2). Where a project needs a minimum viable grant, the first condition fails and a lottery over full awards can beat dividing the same money into grants too small to use. The cost of the randomness is small in our simulations; the cost of the discarded ranking is the one that depends on the field.

Seed grants and lotteries both spread funds more widely than targeting does, and both raise the older question of whether funding should be concentrated among a few researchers or spread across many. The production function does not settle that question: stronger complementarity between capability and resources concentrates the review-informed funder’s grants when the budget is tight in a heavy-tailed field, spreads them when the budget is ample, and has no single direction where capability is evenly spread (Fig. S5; SI Text 3). The production function, the budget, and the field’s capability distribution set the answer jointly.

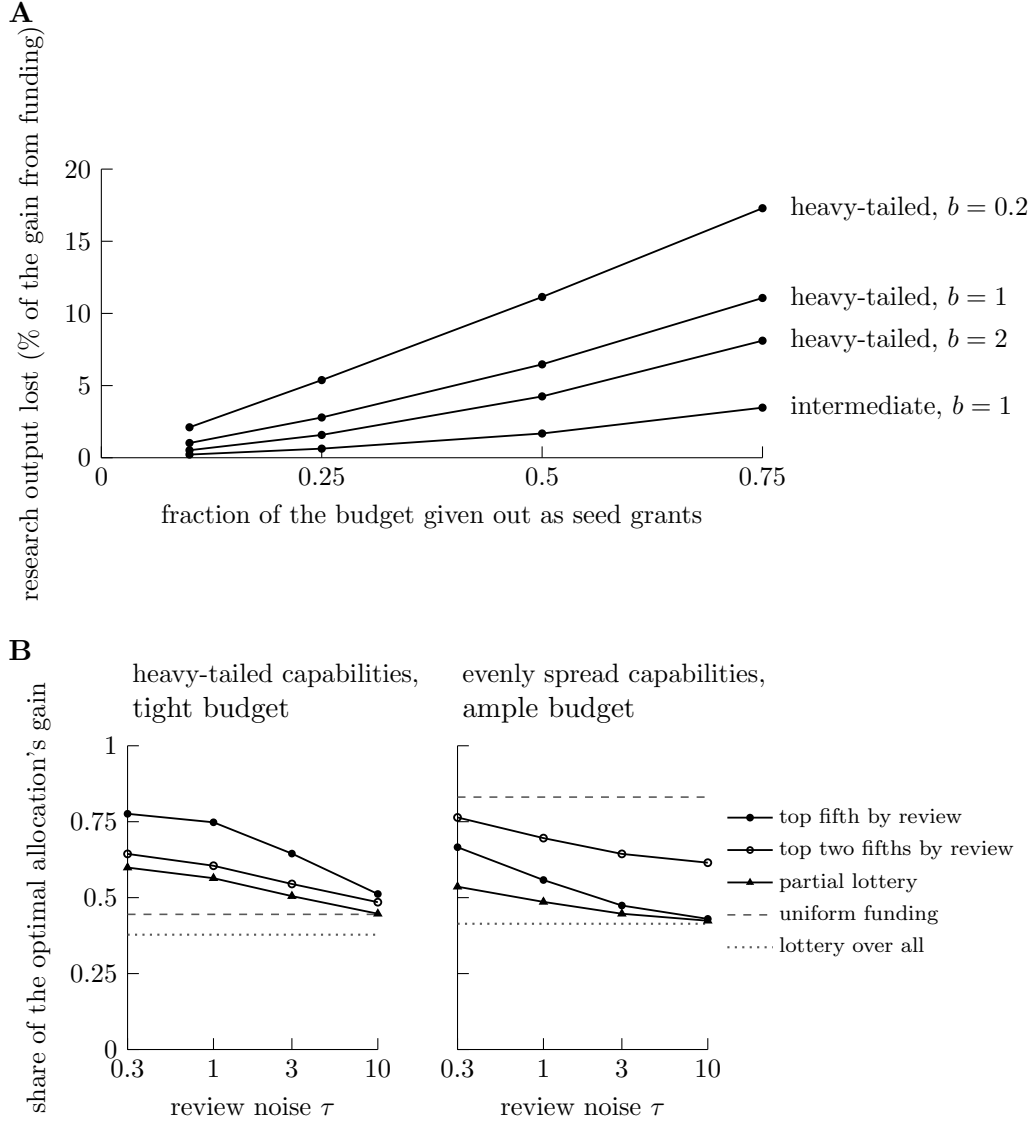


FIGURE 3. Spreading funds. (A) The research output lost to uniform seed grants, as a percent of the research output that the funder's grants add over no funding, when a review-informed funder gives out a fraction of its budget as equal grants to all researchers and targets the rest; the loss grows faster than the fraction, and its size is set by what targeting is worth in the field. (B) What five funding schemes gain, by field: each scheme's gain over no funding as a share of the optimal allocation's gain, as review noise increases, in a heavy-tailed field with a tight budget (left) and an evenly spread field with a large budget (right). The top fifth and top two fifths by review score receive equal division; the partial lottery funds half of the top two fifths at random with full-size grants. Settings: SI Materials and Methods.

DISCUSSION

Our results isolate one quantity as key to optimal grant funding: the value of targeting researchers with the greatest gap between their capabilities and their resources. Two features of a field determine this value. The first is the funder's budget relative to researchers' own resources: where the budget can relieve most researchers' bottlenecks, the optimal allocation and equal division produce nearly the same research output. This saturation holds when capability bounds research output, as under the harmonic mean; under a production function with no such bound, such as Cobb-Douglas, the gap rule still holds but the value of targeting need not vanish as the budget grows (SI Text 4). The second is how unequally the capability-resource gaps are distributed. Across every setting we considered, the value of targeting is close to a function of the inequality of the optimal grants alone (SI Text 5), and what makes the gaps unequal is mainly how unequal capability is; the association between capability and resources works through the same channel, since where the capable already hold the resources the gaps are small and even. Two results are not set by this quantity alone. How widely a review-informed funder should spread its grants depends, in addition, on how strongly capability and resources complement each other (Fig. S5); and the timing of spending over the horizon, analyzed in SI Text 6, adds little research output beside the choice of recipients in every setting we test, within the limits stated there.

Within the model, no funding mechanism is best in every field, so a funder choosing among peer review, seed grants, and lotteries should begin by locating its field on the two features (Fig. 4). The first is observable: a funder's annual awards to a field against the field's research spending from every other source. The second is not, because the gaps depend on capabilities the funder cannot see and on the budget itself. Three observable quantities stand in for it. The inequality of research output in the field is the natural stand-in for the inequality of capability, though where grants are small research output tracks resources rather than capability, so this stand-in leans toward the resource distribution. The dispersion of review scores over the whole applicant pool is a second, with the same caveat about noise. The most direct is the response of research output to grants large enough to relieve the bottleneck (Fig. 2A): a program of substantial pilot grants measures the gaps, whereas small seed grants do not. How existing funding is distributed completes the estimate: where resources already sit with the capable, the gaps are smaller and more even than capability inequality alone would imply.

Fig. 4 places four kinds of funding program on the budget axis by order of magnitude, from 2023 and 2024 budget figures; a round corresponds to a year of awards and the map's budget covers two rounds, so a program whose annual awards are

a fraction r of its field’s other annual resources sits at $b = 2r$ (SI Materials and Methods). A national biomedical agency that supplies a large share of its field’s research support, with annual awards comparable to the field’s other resources, sits at the top of the range; a national science agency that funds its fields more thinly near a tenth of that; a continental excellence council near a few hundredths; and a private foundation funding one field among many at a hundredth or below. Where a field lies on the inequality axis is the estimate each funder must make, so the placements say which regime a funder’s budget puts it in, not where it sits within the regime. The regimes differ. A foundation or council with a small budget in a heavy-tailed field faces the largest value of targeting per dollar: review directed at identifying the few researchers of unusually high capability is worth most, even moderately noisy review recovers most of that value, seed grants give up the most research output, and a lottery among applicants above a review threshold gives up a substantial share of what selection by review gains. An agency whose budget approaches the field’s other resources, in a field where capability is evenly spread, faces a small value of targeting: review of any informativeness adds little, seed grants and lotteries give up little, and the model’s best use of the budget is small grants to many researchers.

Three further implications concern peer review. First, most of review’s value comes from separating the few researchers of unusually high capability from the rest. This suggests that a review system that spends most of its effort ranking the many applications of comparable merit spends that effort where the model locates the least research output. Second, the finding that percentile scores barely predict productivity comes from funded NIH grants with scores in the top fifth (Fang et al. 2016), and the studies that find modest predictive power also use funded grants (Li and Agha 2015); both measure prediction among applicants review has already selected, where the model expects review to add least, so a weak association there is consistent with review of considerable value over the whole applicant pool, as the statistical analysis of these studies by Lindner and Nakamura (2015) also concludes. Third, a funder that treats review as more reliable than it is loses more research output than a funder that treats review as noisier than it is, so a funder uncertain about review’s accuracy should err toward undertrust.

The model holds several things fixed, and its results hold within those limits. There is one funder, and researchers do not respond strategically, so competition among funders, proposal effort, risk choice, and the cost of review are outside the model; the contest models of grant competition cover that ground (Gross and Bergstrom 2019, 2025), and the mechanism-design treatment of grants by Carnehl et al. (2025), in which the funder must elicit information from applicants

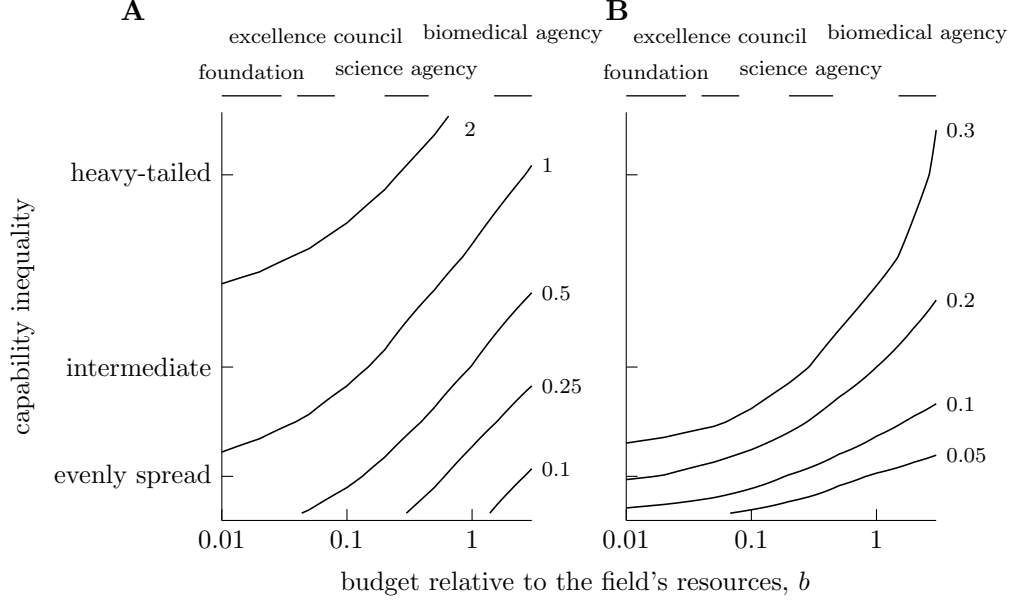


FIGURE 4. Where a funder stands. (A) The value of targeting: the complete-information optimum's gain over uniform funding, as a share of uniform funding's gain over no funding. (B) The value of review at the default noise, as a share of the research output that review-informed funding adds. Both by the funder's budget relative to the field's resources and the inequality of capability, at fixed mean capability; the vertical axis is the Gini coefficient of capability, and its ticks mark the three fields of the paper ($\alpha_K = 1.3, 2$, and 3.5); each contour is labeled by its value. The bars above each panel place four kinds of program on the budget axis by order of magnitude, from 2023 and 2024 annual budget figures with b equal to twice the annual ratio, since the map's budget covers two rounds (SI Materials and Methods); where a field lies on the vertical axis is the estimate each funder must make, and the tail parameters estimated for research productivity lie in its upper part. Two rounds, $\tau_K = 1$, 50 populations per cell; contours interpolated between the 63 cells. Settings: SI Materials and Methods.

who bear the cost of applying, complements a model in which the funder's information arrives for free. Sakamoto (2023) compares review with baseline funding against review with a lottery in a formal model and finds baseline funding better in many situations; our seed grants play the role of baseline funding, and here the comparison turns on the field. Sorzano and Pueche-Granados (2026) also treat funding as a decision under heavy-tailed returns and find more room for lotteries than we do; their heavy tail is in observed impact, whereas ours decomposes research output into unobserved capability and resources, and that decomposition is what gives review its value here. Research output is a single number per round,

so projects do not differ in kind and exploring new directions has no value in the model (Weisberg and Muldoon 2009; Avin 2019b); nor do a researcher’s proposals differ in quality, so this is a model of investigator-level funding. Grants are divisible (Results). The review signal is noisy but unbiased; bias and conservatism in review, which are among the strongest arguments for lotteries, are not modeled (Ginther et al. 2011; Wittteman et al. 2019; Boudreau et al. 2016). Capability is a state rather than a trait, so it holds whatever a researcher has accumulated that raises research output, and doing research adds to it. Capability persists and grows while resources do not persist, so a grant’s lasting effect enters only through capability; durable equipment, data, and infrastructure violate this asymmetry, and the results about compounding and timing (SI Text 6) depend on it while the one-round results do not. The compounding here is of capability through doing research, and differs from the cumulative advantage in which winning a grant raises the chance of winning the next (Merton 1968; Bol et al. 2018). None of the numbers we report is calibrated to a real funder; what applies to a real field is the qualitative features of the dynamics that organize the results.

The model makes four predictions that data on funded research can test. The effect of winning a grant on research output should be larger for recipients whose other resources are small relative to their capability, so a randomized trial of funding, such as the one that found no clear average effect (Barnett et al. 2024), should find its effect concentrated among under-resourced recipients; a null average is what the model predicts where most recipients are already well resourced. Measures that stand in for capability, such as prior record or review score, should predict research output better the larger a grant is relative to the recipient’s other resources. How well review scores predict research output, measured over the whole applicant pool rather than among funded grants, should be higher in fields whose research output distributions have heavier tails. And where funders run lotteries beside review, the research output lost to the lottery should be larger in heavier-tailed fields. The last two predictions compare fields at similar budget scales and review informativeness, and read the tail of research output as a stand-in for the tail of capability.

Our model provides a unifying insight into the grant funder’s challenge. The output-maximizing allocation funds the gap between each researcher’s capability and their resources. Records alone do not pin down that gap, because research output does not show which of the two is scarce; a signal of capability less confounded by resources does, and peer review can provide one. What that signal is worth, and how seed grants and lotteries perform, turn on two features of a field, and the map of those features has regimes in which the answers differ. A funder

deciding among review, lotteries, and seed grants should begin by asking which kind of field it is funding.

MATERIALS AND METHODS

Model. A funder faces n researchers over T funding rounds. Researcher i has capability K_i and resources R_i ; initial values are drawn from Pareto distributions with tail parameters α_K and α_R (smaller values give more unequal fields; below 2 the variance is infinite), coupled through a Gaussian copula with parameter ρ ($\rho = 0.5$ and 0.8 give rank correlations of 0.48 and 0.79 ; a Pearson correlation is undefined when the variances are infinite). In each round, researcher i produces research output drawn from a Poisson distribution with mean $\lambda_i = AK_i R_i / (K_i + R_i)$, proportional to the harmonic mean of the two inputs, where the productivity constant A sets the units of research output (the simulations set $A = 1$). Between rounds capability compounds, $K_i \leftarrow K_i + \epsilon \lambda_i / A$, at rate ϵ . Resources do not accumulate: $R_i = R_{i0} + g_i$, where the baseline R_{i0} recurs every round and the grant g_i is consumed in the round it is given. The unit of analysis is the researcher, not a researcher-project pair. The funder’s total budget over the horizon is $b n \mathbb{E}[R_0]$, so at budget scale $b = 1$ it equals the community’s expected baseline resources for one round; the budget does not depend on the horizon.

Information and strategies. The funder knows the distributions but observes no researcher’s capability or resources. It observes each researcher’s realized research output as it accumulates; a resource signal $R_{i0} + \eta_i$, observed once, with η_i Gaussian with standard deviation τ_R ; and a review signal $K_i + \xi_i$, observed once at the start by the strategies that use it, with ξ_i Gaussian with standard deviation τ_K and independent across researchers, so a reviewer’s misjudgment persists (SI Text 5 redraws review every round). Bayesian strategies hold a posterior over each researcher’s capability and baseline resources, computed by importance sampling with M particles drawn from the prior ($M = 1,000$ for Fig. 2B and C, Fig. 4, and the sensitivity analysis, and 200 elsewhere; convergence in M is checked in SI Materials and Methods), update it after each round, and allocate each dollar to the researcher whose expected research output, given the posterior, rises most with it. The myopic funder maximizes the current round’s expected research output; the forward-looking funder plans the remaining budget over the remaining horizon, treating future research output as equal to its expectation, and re-plans each round (SI Text 6). The complete-information benchmark applies the gap rule to each round’s share of the budget with the true capabilities and resources. Three baselines complete the set: no funding, against which every result is normalized; uniform funding; and track-record funding, which allocates in proportion to the previous round’s research output (SI Materials and Methods, Table S1).

Parameters and scoring. Tail parameters range from 1.3 to 3.5, a range that includes the Pareto tails near 1 to 2 estimated for research productivity (Lotka 1926; Nicholls 1989) and extends well beyond them, since productivity is an imperfect guide to capability (Results); the budget scale b from 0.01 to 3; the compounding rate ϵ from near zero to 0.85; review noise τ_K from 0.05 to 20; the horizon from one round to ten. Default values are $T = 2$, $n = 50$, $\epsilon = 0.1$, $b = 1$, $\alpha_K = \alpha_R = 2$, $\tau_K = \tau_R = 1$, $\rho = 0$. Each strategy is scored by the sum of expected research output along its run, and strategies are compared on identical simulated populations with identical signal and research output draws, so comparisons are paired. Results are reported as shares of a named comparison, most often the research output that funding adds: a strategy’s expected research output minus the expected research output under no funding. Numbers quoted in Results are read from the figure named in the sentence at the stated setting and illustrate a comparison; the claims are the qualitative comparisons, which hold across the ranges of our parameter sweep (SI Text 5). Proofs are in SI Text 1 and SI Text 2; the settings behind each figure are in SI Materials and Methods. Code and data are available in the project repository.

SUPPORTING INFORMATION APPENDIX

The SI Appendix holds what the main text cites without proof or states in one sentence: the model details, strategies, parameter ranges, the particle audit, and the settings behind every figure (SI Materials and Methods); the derivation of the gap rule and its comparison with the sequential optimum (SI Text 1); the lottery proposition (SI Text 2); four extended results with Figs. S2 to S5 (SI Text 3); the robustness of the results to the production function (SI Text 4); the sensitivity of the value of targeting and of review to the parameters the main sweeps hold fixed and to the signal model (SI Text 5); and the timing of funding (SI Text 6). The table below maps each main-text statement to the section that supports it.

Main-text statement	Where in the SI
Model, signals, strategies (Table S1), parameter ranges, copula, particles (Table S2), scoring, settings behind every figure (Table S3), code	SI Materials and Methods
The output-maximizing allocation of a round's budget funds the gap: $g_i^* = \max(cK_i - R_{i0}, 0)$	SI Text 1, Proposition 1
The funded set and every grant increase with the budget; the funding frontier	SI Text 1, Corollary 1
The value of targeting decreases in the budget and vanishes as the budget grows	SI Text 1, Corollary 2; SI Text 3, Fig. S2
Each heuristic can perform worse than uniform division	SI Text 1, Examples 1 and 2, Fig. S1
The round-by-round gap rule is within about one percent of the sequential optimum	SI Text 1, Table S4
Equal division outperforms any lottery over the same pool	SI Text 2, Proposition 2
Targeting's value relative to uniform funding, across budgets and fields	SI Text 3
Overtrusting review loses more research output than undertrusting it	SI Text 3, Fig. S3
Large grants reveal capability; the source of the shift of spending toward later rounds	SI Text 3, Fig. S4
The production function does not settle concentration	SI Text 3, Fig. S5
The ordering of review's value by field holds from Cobb-Douglas to Leontief	SI Text 4, Fig. S6
The ordering of fields by the value of targeting and of review survives changes in mean capability, the capability-resource association, the resource tail, pool size, the resource signal, and the review-signal model; the inequality of the gaps organizes the results	SI Text 5, Tables S6 to S12, Fig. S7
Planning the timing of spending adds little beside choosing recipients, and exists only under complementarity	SI Text 6, Fig. S8; SI Text 4, Table S5

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Model details. Research output in each round is drawn from a Poisson distribution with rate $\lambda_i = AK_i R_i / (K_i + R_i)$, independently across researchers and rounds given the state; the units of research output may be read as publications, findings, or units of scientific value. The production function is proportional to the harmonic mean of capability and resources. The harmonic mean makes the two inputs complements without making either an absolute requirement: a capable

researcher with few resources still produces, by working more slowly or borrowing equipment, but no surplus of one input fully substitutes for a shortage of the other. SI Text 4 repeats the central comparisons for a family of production functions from Cobb-Douglas to Leontief.

Between rounds, capability grows by $\epsilon \Lambda(K_i, R_i)$ with $\Lambda = K_i R_i / (K_i + R_i)$: doing research builds expertise, the same bottleneck that limits research output limits growth, and capability never declines. Growth follows expected research output, so a researcher's accumulation depends on their capability and resources and not on luck in a given round. Resources do not accumulate. The baseline R_{i0} recurs every round and represents the time and support a researcher has independent of the funder; a grant is consumed in the round it is given, and whatever lasts from it (skills, publications, standing) enters the model as capability.

The funder's total budget over the horizon is $b n \mathbb{E}[R_0]$, where $\mathbb{E}[R_0]$ is the expected baseline resources of one researcher, so at $b = 1$ the budget equals the community's expected baseline resources for one round. The budget does not grow with the horizon: a longer horizon varies the timing of spending, never its amount. Strategies that do not plan ahead spend $1/T$ of the budget each round; forward-looking strategies re-plan the entire remaining budget over the remaining horizon each round and execute the current round's allocation. For the timing runs (Fig. S8A and Fig. S4) the budget is scaled against capability instead, $b n \mathbb{E}[K_0]$, so that a community with few resources does not zero the budget. The simulations set the productivity constant $A = 1$; the code parameterizes the budget as $2 b' n \mathbb{E}[R_0]$, so its b' is half of the b reported here.

The funder knows the distributions of capability and resources but observes no researcher's capability or resources directly. It observes each researcher's research output as it accumulates. A resource signal, observed once at the start, gives a noisy estimate of each researcher's baseline resources (noise τ_R), as one might obtain from institutional information. A review signal, also observed once at the start by the strategies that use it, gives a noisy estimate of capability (noise τ_K), as one might obtain from proposal evaluation and expert judgment; the smaller τ_K , the more informative the review. Formally, the resource signal is $R_{i0} + \eta_i$ with $\eta_i \sim N(0, \tau_R^2)$, and the review signal is $K_i + \xi_i$ with $\xi_i \sim N(0, \tau_K^2)$, independent across researchers; both are held fixed over the horizon, so a reviewer's misjudgment of a researcher persists across rounds, and only the research output record accumulates. SI Text 5 reports the variant in which review is redrawn every round, and the variant in which review noise is multiplicative. Realized research output is Poisson with mean λ_i . Bayesian strategies hold a posterior over each researcher's (K_i, R_{i0}) , computed from these likelihoods by importance sampling

Strategy	Information	Planning
No funding	none	none
Uniform	none	none
Track record	track record	none
Myopic, without review	track record + resource signal	current round
Forward, without review	track record + resource signal	remaining horizon
Myopic, with review	track record + resource signal + review	current round
Forward, with review	track record + resource signal + review	remaining horizon

TABLE S1. The funding strategies compared.

with $M = 200$ particles drawn afresh from the prior each round and weighted by the full likelihood of the record and signals to date (the convergence of the results in M is checked below), update it after each round, and allocate each dollar to the researcher whose expected research output, given the posterior, rises most with it, by exact water-filling on the posterior expected marginal values. The forward-looking funder is a certainty-equivalent receding-horizon planner: each round it chooses the current allocation and a plan for the remaining budget to maximize expected research output over the remaining horizon, treating future research output as equal to its expectation, and re-plans after observing the round’s research output; it does not value the information its own grants will generate (SI Text 6). The complete-information benchmark applies the gap rule of Proposition 1 to each round’s equal share of the budget with the true capabilities and resources; SI Text 1 compares it with the exact dynamic optimum.

Strategies. Table S1 lists the strategies compared. The Bayesian strategies differ in information, with or without the review signal, and in planning, myopic (optimizing the current round’s expected research output) or forward-looking (optimizing expected research output over the remaining horizon). Three baselines complete the set: no funding, uniform funding, which splits each round’s budget equally, and track-record funding, which allocates in proportion to the previous round’s research output. Seed grants and lotteries modify these strategies as described in the main text.

Parameter ranges and defaults. The tail parameters α_K and α_R range from 1.3 to 3.5. Empirical distributions of research productivity have Pareto tail parameters near 1 to 2 (Lotka 1926; Nicholls 1989); since productivity is an imperfect guide to capability (main text), the range extends well beyond those estimates toward more equal fields. The budget scale b ranges from 0.01 to 3, from a funder whose budget is small beside the community’s own resources to one able to relieve most researchers’ bottlenecks, with larger budgets in the timing analyses (SI Text 6). The compounding rate ϵ ranges from near zero, where funding raises only current research output, to 0.85, where this round’s funding substantially raises

later capability. The review noise τ_K ranges from 0.05, nearly perfect review, to 20, nearly uninformative review, a range wide enough to contain the noise implied by empirical estimates of review’s predictive accuracy. The copula parameter ρ ranges from -0.5 to 0.8 (rank correlations -0.48 to 0.79), and the horizon T from one round to ten. Capability and resources are coupled through a Gaussian copula: standard normal draws with correlation ρ are mapped through the normal distribution function to uniforms and then through the Pareto quantile functions. A Pearson correlation between two Pareto variables is undefined when either tail parameter is at most 2, so ρ is a copula parameter and not a correlation of the draws. Below a tail parameter of 2 the Pareto distribution has infinite variance, and sample means converge slowly: at $\alpha_K = 1.3$ with the scale set for a mean of 2, the mean capability across 100 populations of 50 is 1.77 and the median population has mean 1.48, because the rare draws that carry the mean are absent from most samples. For that reason the heavy-tailed results are reported with standard errors across populations, and Table S2 reports medians beside means. The rank correlation induced by the copula is $(6/\pi) \arcsin(\rho/2)$. Unless a figure’s settings below say otherwise: $T = 2$; $n = 50$; capability and baseline resources drawn from Pareto distributions with scale 1 and $\alpha_K = \alpha_R = 2$; $\rho = 0$; $\epsilon = 0.1$; $b = 1$; $\tau_K = \tau_R = 1$; seed-grant fraction 0.25 where seed grants are used.

Particles. Bayesian funders compute each researcher’s posterior by importance sampling: M particles (K, R_0) are drawn from the prior afresh in each round and weighted by the full likelihood of the researcher’s record and signals to date. With few particles the posterior can rest on a handful of them, and rare regions of the prior, where the most capable researchers of a heavy-tailed field sit, can be missed. Table S2 checks this. Raising M from 200 to 1,000 raises the value of review by 0.01 to 0.02 in every field at every noise level (the posteriors at $M = 200$ understate the capability of the most capable researchers), and raising it further to 5,000 changes the value of review by at most 0.005, within one standard error everywhere. The value of targeting uses no particles. Fig. 2B and C, the sensitivity analysis of SI Text 5, and Fig. 4 use $M = 1,000$; the remaining simulation figures use $M = 200$, and by the table their values of review are understated by about 0.01 to 0.02 of the no-review shortfall, which does not change any comparison they support.

Scoring and comparison. Each strategy is scored by the sum of expected research output along its run, which equals realized research output averaged over repetitions of the same allocations; this removes simulation noise from comparisons without favoring any strategy. Every configuration is run on independently

TABLE S2. Convergence in the number of particles. The value of review (share of the no-review shortfall recovered; mean over 100 populations, with the median in parentheses) and the effective sample size of the round-one posteriors (median across researchers and populations, with the tenth percentile in parentheses), by field, review noise, and number of particles M . Two rounds, $b = 1$, mean capability 2; the same 100 populations at every M . The value of targeting uses no particles and is unchanged.

α_K	τ_K	$M = 200$	$M = 1000$	$M = 5000$
<i>Value of review, mean (median)</i>				
1.3	0.3	0.91 (0.94)	0.94 (0.96)	0.94 (0.97)
1.3	1	0.77 (0.83)	0.79 (0.84)	0.79 (0.85)
1.3	3	0.48 (0.56)	0.50 (0.58)	0.50 (0.53)
2	0.3	0.84 (0.87)	0.86 (0.89)	0.87 (0.89)
2	1	0.63 (0.67)	0.65 (0.69)	0.65 (0.69)
2	3	0.28 (0.26)	0.30 (0.26)	0.29 (0.27)
3.5	0.3	0.62 (0.65)	0.63 (0.66)	0.64 (0.67)
3.5	1	0.31 (0.32)	0.32 (0.33)	0.32 (0.33)
3.5	3	0.05 (0.04)	0.06 (0.05)	0.06 (0.04)
<i>Effective sample size, median (tenth percentile)</i>				
1.3	0.3	81 (5)	396 (28)	1998 (136)
1.3	1	122 (15)	618 (73)	3081 (369)
1.3	3	142 (36)	711 (164)	3559 (796)
2	0.3	74 (6)	368 (33)	1843 (166)
2	1	124 (18)	616 (90)	3079 (447)
2	3	146 (46)	732 (227)	3667 (1137)
3.5	0.3	92 (12)	460 (56)	2300 (294)
3.5	1	136 (39)	685 (193)	3417 (955)
3.5	3	153 (67)	771 (314)	3857 (1558)

drawn populations, with the same random draws (populations, signals, and realized research output) used for every strategy, so that comparisons between strategies are paired. Results are reported as shares of a named comparison, stated in each figure: most often the research output that funding adds (a strategy’s expected research output minus the expected research output under no funding), and otherwise the no-review shortfall or the optimal allocation’s gain. Standard errors, where shown, are across simulated populations.

Settings by figure. Table S3 lists, for each figure, the settings that differ from the defaults.

Placing programs on the budget axis. A round of the model corresponds to a year of a program’s awards, and the regime map’s budget b is the total over its two rounds, so a program whose annual awards to a field are a fraction r of the field’s annual research spending from every other source sits at $b = 2r$. The

Figure	Settings that differ from the defaults
Fig. 2C	Review noise τ_K from 0.05 to 20; $\alpha_K \in \{1.3, 2, 3.5\}$ with the Pareto scale set to $2(\alpha_K - 1)/\alpha_K$ so that mean capability is 2 in every field; 100 populations per point; $M = 1,000$. The no-review shortfall relative to complete information is 44, 26, and 11 percent of the research output that complete-information funding adds, at the three tail parameters.
Fig. 2B	$T = 20$; 50 populations. No-review is the myopic funder without review; with review is the myopic funder with a single review score observed at the start; $M = 1,000$. The shortfall is the funder's expected research output below the complete-information optimum in that round, as a share of the optimum's gain over no funding. Under heavy-tailed capability ($\alpha_K = 1.3$, mean 2) the no-review shortfall falls from 60 to 14 percent against 6 to 2 percent with review, and the grant correlations run 0.12 to 0.55 against 0.89 in round one.
Fig. S3	True noise $\tau_K = 3$; believed noise from 0.1 to 10; $\alpha_K \in \{1.3, 2\}$; 200 populations. Each negative value is within two standard errors of zero.
Fig. S4	$T = 6$; community with few resources (Pareto scale for resources 0.001); $\alpha_K = 1.3$; no review signal ($\tau_K = 100$); $\epsilon = 10^{-4}$; budget scaled against capability; $b = 1$ (50 populations) and $b = 6$ (24 populations, standard error of each share at most 0.006).
Fig. S8A	$T = 5$; forward-looking funder with review; budget scaled against capability; resource scales 0.001, 0.3, and 3; ϵ from 10^{-4} to 0.85; 200 populations. Five resource scales were run; the figure shows three, and the other two lie between them.
Fig. S8B	T from 2 to 10 (to 5 for $\epsilon \in \{0.05, 0.15, 0.55\}$); $\epsilon \in \{0.05, 0.15, 0.3, 0.55, 0.85\}$; 64 to 200 populations per point. The gain is the forward-looking funder's research output over the myopic funder's, both with review; negative gains, all within noise of zero, are shown as zero.
Fig. 3A	Review-informed forward-looking funder; seed-grant fraction from 0 to 0.75; heavy-tailed field ($\alpha_K = 1.3$) with reliable review ($\tau_K = 0.3$) at $b \in \{0.2, 1, 2\}$, and the intermediate field at $b = 1$; 200 populations. The research output lost is reported as a share of the review-informed funder's gain over no funding in the same setting.
Fig. 3B	Single round; $n = 50$; a fifth of the population funded; review score K_i plus Gaussian noise of standard deviation τ ; expected research output computed exactly given each allocation; 2,000 populations per point (Monte Carlo standard error at most 0.002). Left: $\alpha_K = 1.3$, $b = 0.2$. Right: $\alpha_K = 3.5$, $b = 1$.
Fig. S6, Table S5	Production functions with $\gamma \in \{0, -3\}$ and Leontief; 100 populations per point (50 for Leontief); $T = 2$ for the signal value and $T = 5$ for the timing table. These runs use the greedy allocator, the only one implemented for the general family (the harmonic-mean runs use exact water-filling), so their magnitudes are compared with one another and not with the main-text figures. Leontief allocations use a finer allocation step, with ties at the kink broken by fill order.
Fig. S5	Single round; review-informed funder; γ from 0 to -12 and Leontief; three field-and-budget combinations ($\alpha_K = 1.3$, $b = 0.2$; $\alpha_K = 2$, $b = 2$; $\alpha_K = 3.5$, $b = 2$); 200 populations per point, 400 at the refinement points of the evenly spread curve.
Fig. S7, Tables S6 to S12	$T = 2$; $b = 1$; myopic funders with and without review and the complete-information benchmark; $\rho \in \{-0.5, 0, 0.5, 0.8\}$; $\alpha_R \in \{1.3, 2, 3.5\}$; $n \in \{25, 50, 100, 200\}$; Pareto scale 1 or set to give mean capability 2; review score $K_i + \beta R_{i0}$ plus noise with $\beta \in \{0, 0.25, 0.5, 1\}$; resource signal noise $\tau \in \{0.1, 0.2, 1, 2, 10\}$; and

estimate counts money and not time, and treats each funder's field as the disciplines it mainly funds, so it is an order of magnitude; the bands in the figure span a factor of about two around each placement. United States universities spent \$108.8 billion on research and development in fiscal year 2023, \$55.2 billion of it in the life sciences, and received \$33.1 billion from the Department of Health and Human Services, chiefly the National Institutes of Health, and \$6.7 billion from the National Science Foundation (National Center for Science and Engineering Statistics 2025): a biomedical agency of that size has r near 1.5 (\$33 billion against \$22 billion from other sources), so b near 3, at the top of the map; a science agency of that size funding the other half of academic research has r near 0.15 and b near 0.3. The European Research Council's 16 billion euros over 2021 to 2027 (European Research Council 2025), about 2.3 billion a year, against 86 billion euros of higher-education research spending in the European Union in 2024 (Eurostat 2025), gives r near 0.03 and b near 0.05. A private foundation awarding on the order of \$100 million a year across several fields has r at or below 0.01 in any one of them, and b at or below 0.02.

Code. The model, the sweeps, and the scripts that produce every figure and number in the paper are in the project repository; the archived version will be cited at publication.

SI Text 1. DERIVATION OF THE GAP RULE

The within-round allocation problem. A population of n researchers. Researcher i has capability $K_i > 0$ and baseline resources $R_{i0} \geq 0$, both fixed within the round. A grant $g_i \geq 0$ adds to researcher i 's resources for the round, so that their expected research output is $F(K_i, R_{i0} + g_i)$. We prove the gap rule for every production function of the form

$$F(K, R) = A K h(R/K),$$

where $A > 0$ is the productivity constant and $h: [0, \infty) \rightarrow [0, \infty)$ is continuous, continuously differentiable on $(0, \infty)$, strictly increasing, and strictly concave, so that h' is positive and strictly decreasing. This is the class of production functions with constant returns to scale in (K, R) and diminishing returns to resources. The harmonic mean of the main text is the instance $h(x) = x/(1+x)$, for which $F(K, R) = AKR/(K+R)$; every member of the power-mean family of SI Text 4 with exponent $\gamma < 1$ is an instance, with $h(x) = ((1+x^\gamma)/2)^{1/\gamma}$. Write

$$\varphi_i(g) = A K_i h\left(\frac{R_{i0} + g}{K_i}\right)$$

for researcher i 's expected research output as a function of their grant alone; for the harmonic mean, $\varphi_i(g) = AK_i(R_{i0} + g)/(K_i + R_{i0} + g)$.

Throughout, the round's budget is $B > 0$. Nothing else about the model is used, so the result holds for every realized population and every round's budget; across rounds, the constant c and the funded set are recomputed from the new population state.

The funder observes $(K_i, R_{i0})_{i=1}^n$ and divides the budget so as to maximize expected research output:

$$(P) \quad \max_{g_1, \dots, g_n \geq 0} \sum_{i=1}^n \varphi_i(g_i) \quad \text{subject to} \quad \sum_{i=1}^n g_i \leq B.$$

Three lemmas.

Lemma 1 (the return to a grant). *For each i , on $g \in [0, \infty)$: (i) φ_i is strictly increasing, with $\varphi'_i(g) = Ah'((R_{i0}+g)/K_i) > 0$; (ii) φ_i is strictly concave, since φ'_i is strictly decreasing in g ; (iii) $\varphi_i(g) \uparrow AK_i \bar{h}$ as $g \rightarrow \infty$, where $\bar{h} = \lim_{x \rightarrow \infty} h(x) \leq \infty$. For the harmonic mean, $\varphi'_i(g) = AK_i^2/(K_i+R_{i0}+g)^2$, $\varphi''_i(g) = -2AK_i^2/(K_i+R_{i0}+g)^3$, $\varphi'_i(g) < A$ whenever $R_{i0}+g > 0$, and $\bar{h} = 1$, so expected research output never exceeds AK_i .*

Proof. Differentiating $\varphi_i(g) = AK_i h((R_{i0}+g)/K_i)$ in g gives (i), and (ii) follows because h' is strictly decreasing and its argument increases with g . For (iii), h is increasing, so $h(x) \uparrow \bar{h}$ as $x \rightarrow \infty$. For the harmonic mean, write $S = K_i + R_{i0} + g$; then $\varphi_i(g) = AK_i - AK_i^2/S$, and the stated derivatives follow (note $dS/dg = 1$); $\varphi'_i(g) = AK_i^2/S^2 < A$ exactly when $S > K_i$, that is, when $R_{i0} + g > 0$. $\square$

Lemma 1 says that the marginal value of a dollar to researcher i depends on capability and resources only through their ratio, decreases as that ratio rises, and so increases with capability and decreases with the resources they already hold; that each dollar granted lowers the value of the next; and that, when h is bounded, expected research output never exceeds a ceiling set by capability.

Lemma 2 (existence, uniqueness, exhaustion). *Problem (P) has exactly one solution $g^* = (g_1^*, \dots, g_n^*)$, and it exhausts the budget: $\sum_i g_i^* = B$.*

Proof. The feasible set is compact and convex and the objective $f(g) = \sum_i \varphi_i(g_i)$ is continuous and, by lemma 1(ii), strictly concave, so a maximizer exists and is unique. If $\sum_i g_i^* < B$, increasing any one g_i^* slightly stays feasible and raises f (lemma 1(i)), contradicting optimality. $\square$

Lemma 3 (equal marginal value). *A feasible allocation g with $\sum_i g_i = B$ solves (P) if and only if there is a $\nu > 0$ such that*

$$\varphi'_i(g_i) = \nu \quad \text{for every } i \text{ with } g_i > 0, \quad \varphi'_i(0) \leq \nu \quad \text{for every } i \text{ with } g_i = 0,$$

where $\varphi'_i(0)$ may be $+\infty$ (as it is when $R_{i0} = 0$ and $h'(0^+) = \infty$), in which case researcher i is funded.

Proof. Necessity. Let g^* solve (P). Since $B > 0$ is exhausted (lemma 2), some researcher is funded: $g_i^* > 0$ for some i . Fix any such i and any $j \neq i$, and for $0 < \delta \leq g_i^*$ consider the transfer that replaces g_i^* by $g_i^* - \delta$ and g_j^* by $g_j^* + \delta$, leaving the rest unchanged. The result is feasible, so optimality gives

$$\varphi_j(g_j^* + \delta) - \varphi_j(g_j^*) \leq \varphi_i(g_i^*) - \varphi_i(g_i^* - \delta).$$

Dividing by δ and letting $\delta \downarrow 0$ yields $\varphi'_j(g_j^*) \leq \varphi'_i(g_i^*)$. If j is also funded, the same argument with i and j exchanged gives the reverse inequality, so all funded researchers share a common marginal value; call it $\nu = \varphi'_i(g_i^*) > 0$ (positivity by lemma 1(i)). For unfunded j , the displayed inequality reads $\varphi'_j(0) \leq \nu$.

Sufficiency. Let g satisfy the condition with multiplier ν , and let $\tilde{g}$ be any feasible allocation. Concavity of φ_i gives, for every i , $\varphi_i(\tilde{g}_i) \leq \varphi_i(g_i) + \varphi'_i(g_i)(\tilde{g}_i - g_i)$. For funded i , $\varphi'_i(g_i) = \nu$, so $\varphi'_i(g_i)(\tilde{g}_i - g_i) = \nu(\tilde{g}_i - g_i)$. For unfunded i , $g_i = 0$ and $\tilde{g}_i - g_i = \tilde{g}_i \geq 0$, so $\varphi'_i(0)(\tilde{g}_i - g_i) \leq \nu(\tilde{g}_i - g_i)$. Summing over i :

$$f(\tilde{g}) \leq f(g) + \nu \sum_i (\tilde{g}_i - g_i) = f(g) + \nu \left(\sum_i \tilde{g}_i - B \right) \leq f(g),$$

since $\tilde{g}$ is feasible and $\nu > 0$. So g solves (P). $\square$

Lemma 3 is the formal content of the sentence in the main text: an allocation is optimal exactly when no dollar can be moved to a researcher who can produce more value with it.

The proposition.

Proposition 1 (the gap rule). *Let $F(K, R) = AK h(R/K)$ with h increasing and strictly concave. Then problem (P) has the unique solution*

$$g_i^* = \max(cK_i - R_{i0}, 0), \quad i = 1, \dots, n,$$

where $c > 0$ is the unique constant satisfying the budget identity $\sum_{i=1}^n \max(cK_i - R_{i0}, 0) = B$. Equivalently, $c = (h')^{-1}(\nu/A)$, where ν is the common marginal value of lemma 3; for the harmonic mean, $c = \sqrt{A/\nu} - 1$ with $\nu \in (0, A)$.

Proof. Let g^* be the unique solution (lemma 2) and ν the multiplier that lemma 3 associates with it. The proof proceeds in four steps.

Step 1: $c = (h')^{-1}(\nu/A)$ is well defined and positive. Some researcher i is funded, and for them lemma 3 gives $\nu = \varphi'_i(g_i^*) = Ah'((R_{i0} + g_i^*)/K_i)$, so ν/A lies in the range of h' ; since h' is strictly decreasing, its inverse is defined there, and

$c = (h')^{-1}(\nu/A) = (R_{i0} + g_i^*)/K_i > 0$ because $g_i^* > 0$. For the harmonic mean, $h'(x) = 1/(1+x)^2$, so $(h')^{-1}(y) = 1/\sqrt{y} - 1$ and $c = \sqrt{A/\nu} - 1$; positivity of c is then $\nu < A$, which lemma 1 gives directly.

Step 2: funded researchers receive exactly their gap. Let $g_i^* > 0$. By lemma 3, $A h'((R_{i0} + g_i^*)/K_i) = \nu$, so $(R_{i0} + g_i^*)/K_i = (h')^{-1}(\nu/A) = c$ and $g_i^* = c K_i - R_{i0}$. Since $g_i^* > 0$, this equals $\max(c K_i - R_{i0}, 0)$. Every funded researcher is thus brought to the same ratio of resources to capability, c .

Step 3: unfunded researchers have no gap. Let $g_i^* = 0$. By lemma 3, $A h'(R_{i0}/K_i) = \varphi'_i(0) \leq \nu = A h'(c)$, and since h' is strictly decreasing, $R_{i0}/K_i \geq c$, that is, $R_{i0} \geq c K_i$. Then $\max(c K_i - R_{i0}, 0) = 0 = g_i^*$.

Steps 2 and 3 establish $g_i^* = \max(c K_i - R_{i0}, 0)$ for every i , and the budget identity is exhaustion (lemma 2). It remains to show c is the only constant satisfying that identity.

Step 4: the budget identity pins c uniquely. Define $G(c) = \sum_{i=1}^n \max(c K_i - R_{i0}, 0)$ for $c \geq 0$. G is continuous and nondecreasing, with $G(0) = 0$ and $G(c) \rightarrow \infty$ as $c \rightarrow \infty$. Moreover G is strictly increasing wherever it is positive: if $G(c) > 0$, then $c K_j > R_{j0}$ for some j , and for any $c' > c$, $G(c') \geq G(c) + K_j (c' - c) > G(c)$. Now suppose $c_1 < c_2$ both satisfy $G(c) = B$. Since $B > 0$, $G(c_1) > 0$, so $G(c_2) > G(c_1) = B$, a contradiction. $\square$

The proof uses only that the marginal value of a dollar depends on K and R through the ratio R/K and falls as that ratio rises. Constant returns to scale give the first property and diminishing returns to resources the second; the harmonic form contributes nothing beyond the closed form of c . The constant c is the ratio of resources to capability at which the optimal allocation leaves every funded researcher, and so also the rate at which the rule converts a unit of capability into a target level of resources.

Consequences.

Corollary 1 (funding frontier and budget comparative statics). *(i) Researcher i is funded, $g_i^* > 0$, exactly when $K_i > R_{i0}/c$. (ii) As a function of the budget B , the constant $c(B)$ is continuous and strictly increasing; hence the funded set is nondecreasing in B , every funded researcher's grant $c(B) K_i - R_{i0}$ is strictly increasing in B , and the common marginal value $\nu(B) = A h'(c(B))$ is strictly decreasing in B . Both parts follow from the formula and Step 4.*

Corollary 2 (the value of targeting vanishes in the budget). *Let $f^*(B)$ denote the optimal value of (P) and $f^u(B) = \sum_i \varphi_i(B/n)$ the value of uniform funding. Then*

$0 \leq f^*(B) - f^u(B)$ for every B . If h is bounded, $\bar{h} < \infty$, then $f^*(B) - f^u(B) \rightarrow 0$ as $B \rightarrow \infty$.

Proof. The first inequality holds because uniform funding is feasible. For the second: by lemma 1(iii), $\varphi_i(g) < AK_i \bar{h}$ for every g , so $f^*(B) \leq A\bar{h} \sum_i K_i$; and $f^u(B) = \sum_i \varphi_i(B/n) \rightarrow A\bar{h} \sum_i K_i$ as $B \rightarrow \infty$. Hence $0 \leq f^*(B) - f^u(B) \leq A\bar{h} \sum_i K_i - f^u(B) \rightarrow 0$. $\square$

Two remarks. First, the corollary says nothing about monotonicity: $f^*(B) - f^u(B)$ is zero at $B = 0$ and positive for small $B > 0$ whenever researchers differ, so it cannot decrease throughout. The decrease reported in the main text is a numerical finding about the ratio $(f^* - f^u)/(f^u - f^0)$, with f^0 the research output under no funding: it fell monotonically in the budget, on budget scales from 0.02 to 3, in every one of 300 populations of 50 researchers drawn from the default distributions, and in the population of Fig. S2. Since uniform funding's own gain over no funding does not vanish (it approaches $A\bar{h} \sum_i K_i - \sum_i \varphi_i(0) > 0$), the ratio vanishes with the difference. Second, boundedness of h is what makes targeting's value vanish: in the power-mean family, h is bounded exactly when $\gamma < 0$, with $\bar{h} = 2^{-1/\gamma}$ (so $\bar{h} = 1$ for the harmonic mean), whereas under Cobb-Douglas production ($\gamma \rightarrow 0$, $h(x) = \sqrt{x}$) research output has no ceiling set by capability and $f^*(B) - f^u(B)$ grows without bound in B for any population with unequal capabilities. The Leontief form $h(x) = \min(1, x)$ is the limit $\gamma \rightarrow -\infty$ of the family; it is not strictly concave, so the proposition does not apply to it directly, and it is treated numerically in SI Text 4.

Example 1 (funding the track record can produce less than uniform funding). *Two researchers, harmonic production, $A = 1$, budget $B = 2$. Researcher 1: $K_1 = 8$, $R_{10} = 8$ (productive and well resourced; expected research output $\varphi_1(0) = 4$). Researcher 2: $K_2 = 4$, $R_{20} = 1$ (capable but with few resources; $\varphi_2(0) = 4/5$). Allocating in proportion to expected research output gives $g = (5/3, 1/3)$ and total research output $285/53 \approx 5.38$. Uniform funding, $g = (1, 1)$, gives $284/51 \approx 5.57$. The gap rule ($c = 3/4$) gives $g^* = (0, 2)$ and $40/7 \approx 5.71$.*

Example 2 (funding the under-resourced can produce less than uniform funding). *Two researchers, harmonic production, $A = 1$, budget $B = 2$. Researcher 1: $K_1 = 10$, $R_{10} = 5$. Researcher 2: $K_2 = 1/2$, $R_{20} = 0$ (the scarcest resources in the population). Directing the budget to the researcher with the fewest resources gives total research output $56/15 \approx 3.73$. Uniform funding gives $49/12 \approx 4.08$. The gap rule ($c = 2/3$) gives $g^* = (5/3, 1/3)$ and $21/5 = 4.2$.*

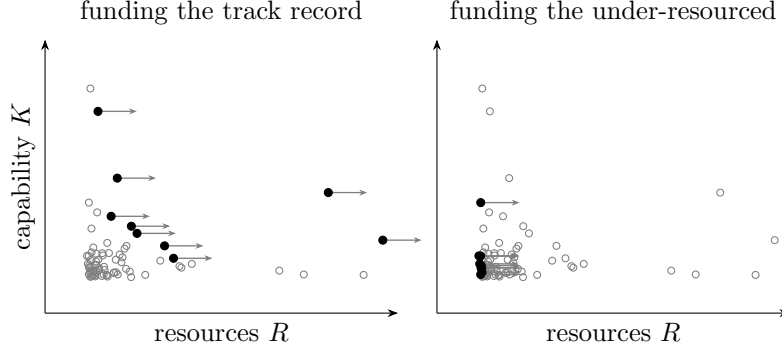


FIGURE S1. The two intuitive schemes on the population of Fig. 1, with the budget of that figure’s smaller frontier divided equally among nine researchers. Left: funding the track record gives the grants (arrows) to the nine researchers with the highest expected research output; most of them are well resourced and of mid-dling capability, so an added dollar adds little. Right: funding the under-resourced gives the grants to the nine researchers with the fewest resources, chosen without regard to capability, so most of the money goes to researchers of low capability. On this population the gap rule adds 4.0 units of expected research output over no funding; funding the track record adds 2.0, funding the under-resourced 2.1, and dividing the budget uniformly across all 75 researchers 2.0.

The one-round rule and the sequential optimum. Proposition 1 maximizes one round’s expected research output. Over T rounds a grant also raises the recipient’s later capability, by $\epsilon \Lambda(K_i, R_{i0} + g_i)$ per round, so the allocation that maximizes expected research output over the horizon can differ from the gap rule in two ways: in whom it funds within a round, since a dollar’s future value depends on the recipient’s growth, and in how it divides the budget across rounds. Because capability growth follows expected research output, the complete-information problem over T rounds is a deterministic program. For $T = 2$ we solve it to global optimality: the round-1 grants and the round-1 share of the budget are chosen freely (a softmax parameterization over the n recipients and a logistic share, optimized by quasi-Newton iteration from the gap rule as the starting point), and the remaining budget is allocated in round 2 by the gap rule on the grown capabilities, which is optimal for the last round. The two-round objective is concave in the grants, so the stationary point the iteration reaches is the global optimum up to numerical tolerance: the harmonic mean $\Lambda(K, R)$ is jointly concave and increasing in (K, R) ; the round-2 value of a capability vector, the maximum of $\sum_i \Lambda(K'_i, R_{i0} + g_{i2})$ over the round-2 grants, is therefore concave and increasing in K' , because partial maximization of a jointly concave function preserves concavity; and each grown

capability $K'_i = K_i + \epsilon \Lambda(K_i, R_{i0} + g_{i1})/A$ is concave in g_{i1} , so the composition is concave in the round-1 grants. Table S4 compares this optimum with the benchmark used throughout the paper, the gap rule applied to half the budget in each round, on 20 populations of 50 at each setting, with mean capability 2. For $T = 5$ we restrict the comparison to the division of the budget across rounds, with the gap rule choosing recipients in each round, and compare the best division with the even one.

The benchmark loses at most 1.4 percent of the research output that funding adds in any population, and under 1 percent on average in every setting; its grants correlate with the optimum's at 0.98 or above. The loss grows with the compounding rate and with the budget, is larger where capability is evenly spread, and is indistinguishable from zero at the default compounding rate. The two-round optimum spends 36 to 49 percent of the budget in the first round, because the capability that round-1 research output builds makes round-2 dollars more productive; the same shift appears in the forward-looking funder of SI Text 6. Over five rounds, with the gap rule choosing recipients, the even division of the budget across rounds loses at most 4.0 percent of what the best division adds at the highest compounding rate, and at most 0.1 percent at the default rate, and the best division shifts spending later (center of mass 0.51 to 0.65). The one-round rule is therefore the right benchmark for the sequential problem at the compounding rates we study, and the value of information reported in the main text is measured against a benchmark within about one percent of the sequential optimum.

SI Text 2. LOTTERIES AND EQUAL DIVISION

Fix a round, a budget $B > 0$, and a pool $P \subseteq \{1, \dots, n\}$ of m researchers. The *equal division* over P grants each researcher in P the amount B/m . A *lottery* over P with $k \leq m$ winners selects k researchers from P uniformly at random and grants each winner B/k . Expected research output is evaluated as in SI Text 1: given an allocation g , it is $\sum_i \varphi_i(g_i)$, and for a random allocation, the expectation is taken over the draw as well.

Proposition 2 (lotteries are outperformed by equal division). *For every pool P , budget B , and winner count $k \leq m$, the lottery's expected research output does not exceed the equal division's, with equality only if $k = m$. In particular, a lottery over the whole population, at any winner count, produces at most uniform funding's expected research output.*

Proof. Researchers outside P receive nothing under either scheme. For $i \in P$, the lottery's grant to i is a random variable $\tilde{g}_i$ equal to B/k with probability k/m

TABLE S4. The round-by-round gap rule against the sequential optimum under complete information. Two rounds: the research output lost by applying the gap rule to half the budget in each round, as a percent of the research output that the exact two-round optimum adds over no funding (mean and maximum over 20 populations); the correlation between the two allocations' first-round grants (mean and minimum); and the share of the budget the optimum spends in round one. Five rounds: the research output lost by an even division of the budget across rounds, as a percent of what the best division adds, with the gap rule choosing recipients in every round, and the center of mass of the best division (0.5 is even; higher is later). Mean capability 2; $n = 50$.

α_K	ϵ	b	two rounds, optimum			five rounds, division only		
			loss, % (max)	correlation (min)	round-1 share	even division loses, %	center of mass	
1.3	0.1	0.2	0.0 (0.0)	1.000 (1.000)	0.49	0.0	0.51	
1.3	0.1	1	0.0 (0.0)	1.000 (1.000)	0.49	0.1	0.51	
1.3	0.1	2	0.0 (0.0)	1.000 (1.000)	0.49	0.1	0.51	
1.3	0.5	0.2	0.1 (0.4)	0.999 (0.997)	0.46	0.4	0.54	
1.3	0.5	1	0.2 (0.6)	1.000 (0.999)	0.45	1.0	0.55	
1.3	0.5	2	0.3 (0.5)	1.000 (0.999)	0.45	1.4	0.55	
1.3	0.85	0.2	0.3 (1.1)	0.999 (0.994)	0.44	0.9	0.56	
1.3	0.85	1	0.6 (1.4)	0.999 (0.998)	0.42	2.1	0.57	
1.3	0.85	2	0.7 (1.2)	1.000 (0.997)	0.42	2.9	0.58	
2	0.1	0.2	0.0 (0.0)	1.000 (1.000)	0.49	0.0	0.51	
2	0.1	1	0.0 (0.0)	1.000 (1.000)	0.49	0.1	0.52	
2	0.1	2	0.0 (0.0)	1.000 (1.000)	0.49	0.1	0.52	
2	0.5	0.2	0.2 (0.4)	0.999 (0.997)	0.45	0.6	0.55	
2	0.5	1	0.3 (0.5)	1.000 (0.999)	0.44	1.2	0.57	
2	0.5	2	0.3 (0.4)	1.000 (0.999)	0.44	1.6	0.57	
2	0.85	0.2	0.4 (1.1)	0.997 (0.989)	0.41	1.2	0.58	
2	0.85	1	0.8 (1.2)	0.999 (0.996)	0.41	2.6	0.60	
2	0.85	2	0.8 (1.1)	0.999 (0.997)	0.41	3.3	0.60	
3.5	0.1	0.2	0.0 (0.0)	1.000 (0.999)	0.48	0.1	0.52	
3.5	0.1	1	0.0 (0.0)	1.000 (1.000)	0.49	0.1	0.52	
3.5	0.1	2	0.0 (0.0)	1.000 (1.000)	0.49	0.1	0.52	
3.5	0.5	0.2	0.3 (0.5)	0.997 (0.994)	0.41	1.3	0.60	
3.5	0.5	1	0.4 (0.4)	0.999 (0.998)	0.43	1.8	0.59	
3.5	0.5	2	0.4 (0.4)	0.999 (0.999)	0.44	1.9	0.58	
3.5	0.85	0.2	0.8 (1.2)	0.993 (0.985)	0.36	2.8	0.65	
3.5	0.85	1	0.9 (1.1)	0.997 (0.995)	0.39	3.7	0.63	
3.5	0.85	2	0.9 (1.0)	0.998 (0.997)	0.41	4.0	0.62	

and 0 otherwise, so $\mathbb{E}[\tilde{g}_i] = B/m$. By linearity, the lottery's expected research output is $\sum_{i \notin P} \varphi_i(0) + \sum_{i \in P} \mathbb{E}[\varphi_i(\tilde{g}_i)]$. Since φ_i is strictly concave (Lemma 1(ii)), Jensen's inequality gives $\mathbb{E}[\varphi_i(\tilde{g}_i)] \leq \varphi_i(\mathbb{E}[\tilde{g}_i]) = \varphi_i(B/m)$ for every $i \in P$, with equality only if $\tilde{g}_i$ is degenerate, that is, $k = m$. Summing over i yields the claim; taking $P = \{1, \dots, n\}$ yields the comparison with uniform funding. $\square$

Remark 1 (what the proposition does and does not cover). *The proposition compares expected research output within a fixed pool. It measures the effect of the randomization alone: whatever pool review selects, replacing the lottery within that pool by the equal division of the same money weakly increases expected research output. The proposition says nothing about the choice of pool, about the review ranking within the pool that the equal division also ignores, or about anything a lottery saves outside the model (review costs, proposal effort, bias).*

Remark 2 (minimum viable grant size). *The proposition needs divisible grants and diminishing returns at every grant size. Suppose instead that a project produces nothing unless its grant reaches a minimum m , and that $B < m|P|$ for a pool P . Equal division then gives every researcher in P less than m and produces no research output, while a lottery with $k = \lfloor B/m \rfloor$ winners, each receiving $B/k \geq m$, produces the research output of k funded projects. Where minimum project scale binds, the comparison of the proposition reverses, and this qualification applies wherever the main text compares lotteries with equal division.*

SI Text 3. EXTENDED RESULTS

The value of targeting decreases in the budget. The value of targeting is the expected research output the optimal allocation adds over equal division of the same budget. By Corollary 2 of SI Text 1 it vanishes as the budget grows, for every production function under which research output has a ceiling set by capability; that it also decreases throughout is a numerical finding, which held in every population we sampled (SI Text 1). Fig. S2 shows it for one population, as a fraction of uniform funding’s gain over no funding. The same dependence on the budget sets how much research output seed grants and lotteries give up and when the informed funder concentrates rather than spreads its grants (main text).

For a funder allocating with records and review rather than with complete information, the value of targeting relative to uniform funding’s gain over no funding decreases from 1.17 to 0.72 across budget scales $b = 0.2$ to 2 with heavy-tailed capability ($\alpha_K = 1.3$) and reliable review ($\tau_K = 0.3$), and from 0.74 to 0.25 in the intermediate field with the default signal ($\alpha_K = 2$, $\tau_K = 1$); two rounds, 200 simulated populations per point, the populations of Fig. 4A. Giving up targeting loses little research output under the same conditions that make review worth little: where capability is evenly spread, the budget is large, or review is uninformative.

Overtrusting review. The results on review’s value require the funder to weight review at its true informativeness. A funder that overtrusts review, treating a noisy signal as reliable, can do worse than one that ignores review altogether: with moderately spread capability, overtrust turns review’s value negative; with

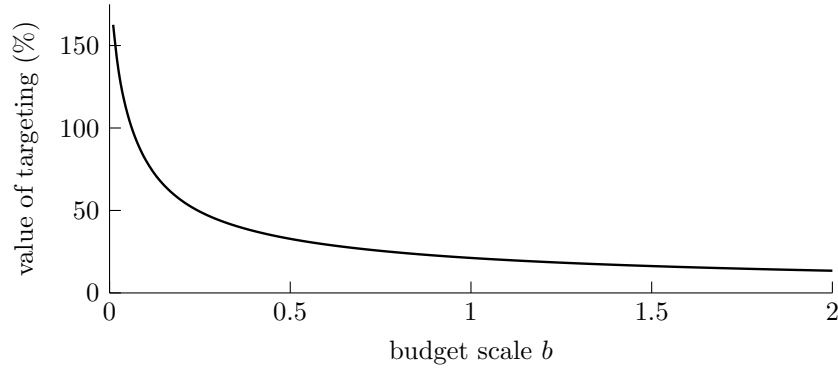


FIGURE S2. The value of targeting decreases in the budget. The additional expected research output of the optimal allocation over uniform funding, as a fraction of uniform funding’s gain over no funding, against the budget scale b , for one population drawn from the default distributions. Corollary 2 proves that the value vanishes as the budget grows; the decrease throughout is a numerical finding (SI Text 1).

heavy-tailed capability it reduces the value without reversing it (Fig. S3). A funder that overtrusts tenfold (true $\tau_K = 3$, belief 0.3) loses more than review’s full value at the intermediate capability distribution, whereas a funder that undertrusts tenfold (true $\tau_K = 0.3$, belief 3) retains over half of it. Overtrusting review loses more research output than undertrusting it.

Grant size, learning, and the schedule. How much a funder can learn from its own grants depends on their size. A researcher on a small grant produces research output nearly proportional to the grant and nearly independent of capability, so a funder whose grants are small learns almost nothing it can exploit later: in a field whose researchers have few resources, with no review signal, such a funder gains almost nothing over uniform funding. As grants approach the scale of capability itself, research output separates the capable from the rest, and the funder recovers most of the value of discrimination, with timing that spends a small share early, so that research output can be observed, and the rest once the funder has learned who is capable (Fig. S4). In a community whose researchers have few resources, with heavy-tailed capability and no review signal, over six rounds, the Bayesian funder’s gain over uniform funding increases from a third of one percent of uniform funding’s research output at the default grant size to nine percent at six times that size and ten percent at twelve times, roughly the gain a reliable review signal delivers at the default size.

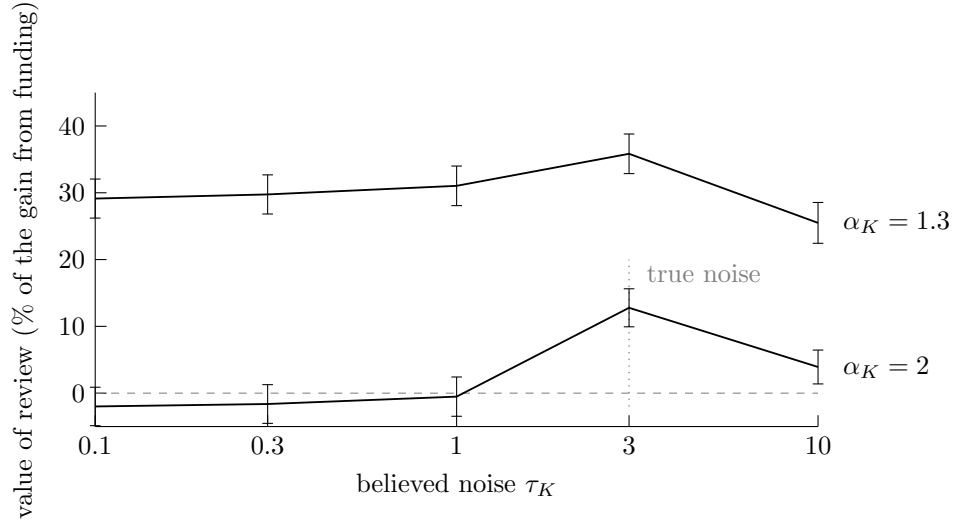


FIGURE S3. Overtrusting review loses more research output than undertrusting it. The value of review as a percent of the research output that a correctly calibrated review-informed funder adds over no funding, as the funder’s believed signal noise varies while the true noise is $\tau_K = 3$ (dotted rule); bars are one standard error. At the intermediate capability distribution, overtrust takes the value below zero. Settings: SI Materials and Methods.

The shift of spending toward later rounds comes from the research output researchers produce whether or not they are funded: that research output compounds and reveals capability while the funder waits. Growth from grant-added research output alone would shift spending slightly toward earlier rounds, since earlier grants have longer to compound. With growth from unfunded research output switched off, the center of mass of spending is 0.48 to 0.49, below even; with growth from grant-added research output switched off, it is 0.53 to 0.68 (five rounds).

Concentration and the production function. One might hope that the production function settles whether funding should be concentrated among a few researchers or spread across many: the harder it is to substitute resources for capability, the more the money should go to the few researchers who can use it. Across the family of production functions of SI Text 4, from Cobb-Douglas to Leontief, it does not (Fig. S5). With a tight budget in a heavy-tailed field, stronger complementarity does concentrate the review-informed funder’s grants: the Gini coefficient of its grants rises from 0.92 to 0.95. With an ample budget, stronger complementarity spreads them, from 0.35 to 0.21 in the intermediate field and from 0.63 to 0.45 in a heavy-tailed field, a pattern consistent with a funder that, once the budget allows, brings every researcher’s resources up toward the level

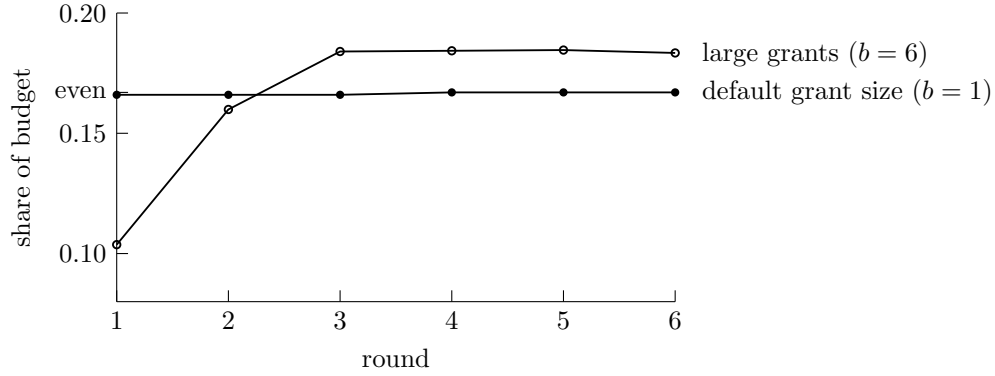


FIGURE S4. With large grants the funder spends a small share first and the rest once it has observed research output. The share of the budget spent in each of six rounds, in a field whose researchers have few resources, with heavy-tailed capability and no review signal, at the default grant size (filled) and at six times that size (open). Settings: SI Materials and Methods.

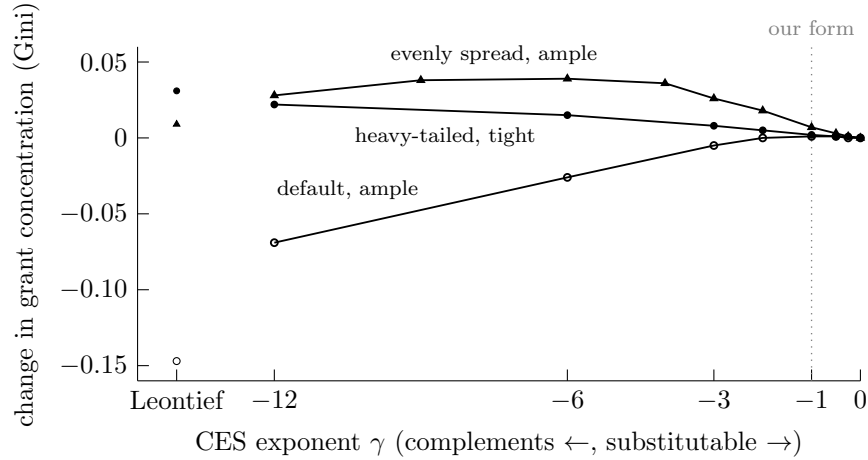


FIGURE S5. Complementarity concentrates funding only when the budget is small. The change in the Gini coefficient of the review-informed funder's grants, relative to its Cobb-Douglas level, as the production function moves from Cobb-Douglas ($\gamma = 0$) toward Leontief (detached marks at the left edge), for three field-and-budget combinations. Settings: SI Materials and Methods.

their capability can use. Where capability is evenly spread and the budget ample, the relation is not even monotone: the Gini coefficient is 0.16 at Cobb-Douglas, peaks at 0.196 near $\gamma = -6$, and falls to 0.185 at $\gamma = -12$, 3.5 standard errors above the $\gamma = -12$ level. How widely funds should be spread is set jointly by the production function, the budget, and the field's capability distribution; no one of the three decides it alone.

SI Text 4. ROBUSTNESS TO THE PRODUCTION FUNCTION

This appendix asks which of our results depend on the harmonic-mean production function of SI Materials and Methods. We embed that function in a family indexed by an exponent γ : expected research output is $A M_\gamma(K, R)$, where $M_\gamma(K, R) = ((K^\gamma + R^\gamma)/2)^{1/\gamma}$ is the power mean of capability and resources. At $\gamma = -1$ the power mean is the harmonic mean $2KR/(K+R)$, which is our production function up to the constant A . In the limit $\gamma \rightarrow 0$ (Cobb-Douglas) it is the geometric mean $\sqrt{KR}$, and a shortfall in either input can be offset by more of the other; as γ decreases the inputs grow harder to substitute; in the limit $\gamma \rightarrow -\infty$ (Leontief) it is $\min(K, R)$, and research output depends only on the scarcer input. We re-run the model's central comparisons across this family. Settings: SI Materials and Methods.

The value of the review signal survives the whole family (Fig. S6). At every production function tested, from Cobb-Douglas to Leontief, the signal's value increases as capability grows more heavy-tailed and as review grows more informative. The ordering of review's value by field (main text) is therefore not an artifact of our choice of production function. What changes with the production function is the scale. The more complementary the inputs, the more review is worth: under heavy-tailed capability with reliable review, the signal's value increases from about 16 percent of the research output that funding with review adds at Cobb-Douglas to about 47 percent at $\gamma = -3$ and about half at Leontief (15.8, 47.1, and 51.4 percent). Cobb-Douglas lies below the two more complementary production functions at every tested tail and noise level at which the value is distinguishable from zero; $\gamma = -3$ and Leontief are within one standard error of each other at the noisiest review under heavy tails. Complementarity raises what targeting is worth, and with it the value of the information that targeting requires.

The timing results of SI Text 6 do not survive the substitutable end of the family (Table S5). Under Cobb-Douglas production the gain from planning ahead is indistinguishable from zero at every compounding rate, and the forward-looking funder's timing stays within about a hundredth of even. At $\gamma = -3$, a production function more complementary than ours, both the gain from planning ahead and the shift of spending toward later rounds increase with the compounding rate and exceed their Cobb-Douglas values at every rate. The timing results hold where capability and resources are complements and vanish where they are substitutable.

How widely the informed funder spreads its grants across this family is reported in the main text (Fig. S5).

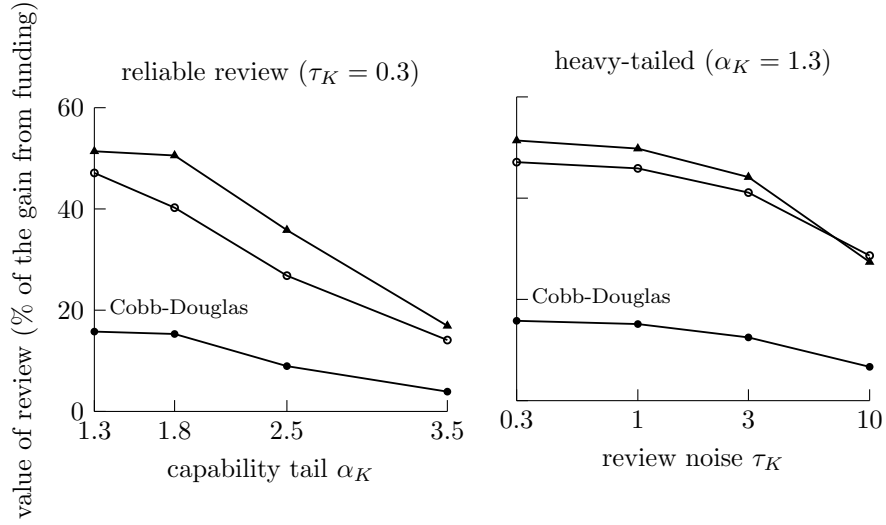


FIGURE S6. The structure of review's value survives the production function. The value of the review signal, as a percent of the research output that funding with review adds over no funding, across the family of production functions: at every production function tested, the value increases as capability grows more heavy-tailed (left) and as review grows more informative (right), and its scale increases with complementarity. Filled circles: Cobb-Douglas; open circles: $\gamma = -3$; triangles: Leontief. Settings: SI Materials and Methods.

SI Text 5. SENSITIVITY OF THE VALUE OF TARGETING AND OF REVIEW

The main text varies the tail of the capability distribution, the budget, and review noise, and holds fixed the mean of the capability distribution, the association between capability and resources ($\rho = 0$), the tail of the resource distribution ($\alpha_R = 2$), the size of the applicant pool ($n = 50$), the noise of the resource signal ($\tau_R = 1$), and the form of the review signal (capability plus additive Gaussian noise, observed once). This appendix varies each of these and asks two questions: whether the ordering of fields by the value of targeting and of review survives, and what single quantity, if any, organizes the results. Every run uses two rounds, the default budget $b = 1$, the myopic funders with and without review, the complete-information benchmark, and $M = 1,000$ particles; 50 simulated populations per setting, 100 in the noise sweeps. The value of targeting is the complete-information optimum's gain over uniform funding as a share of uniform funding's gain over no funding; the value of review is the share of the no-review shortfall that the review signal recovers, as in Fig. 2C.

Mean capability. A Pareto distribution with a fixed scale has mean $\alpha/(\alpha - 1)$, so at scale 1 the heavy-tailed field ($\alpha_K = 1.3$) has mean capability 4.3, the

TABLE S5. The gain from planning ahead and the forward-looking funder’s timing, by production function. Five rounds. The gain from planning ahead is the forward-looking funder’s gain over the myopic funder, both with review, as a percent of the research output that the forward-looking funder’s funding adds; the center of mass of spending is 0.5 when spending is even and larger when spending shifts toward later rounds. One hundred simulated populations per point; standard errors of the gain are at most 0.01 (Cobb-Douglas) and 0.4 ($\gamma = -3$) percentage points.

compounding rate ϵ	0.05	0.15	0.30	0.55	0.85
<i>Value of timing (percent of the gain from funding)</i>					
Cobb-Douglas ($\gamma = 0$)	-0.01	-0.01	0.00	0.01	0.03
$\gamma = -3$	0.05	0.56	1.96	4.56	6.52
<i>Center of mass of optimal spending</i>					
Cobb-Douglas ($\gamma = 0$)	0.499	0.500	0.501	0.505	0.512
$\gamma = -3$	0.511	0.540	0.577	0.623	0.660

TABLE S6. The value of review by field at a fixed Pareto scale and at a fixed mean capability. Share of the no-review shortfall recovered, at review noise $\tau_K = 0.3, 1$, and 3; 100 simulated populations per entry; $M = 1,000$ particles. The mean capability shown is the realized mean over the populations.

α_K	mean K	fixed scale 1			fixed mean 2			
		$\tau_K = 0.3$	1	3	mean K	$\tau_K = 0.3$	1	3
1.3	3.84	0.96	0.90	0.70	1.77	0.94	0.79	0.50
2	1.97	0.86	0.65	0.30	1.97	0.86	0.65	0.30
3.5	1.39	0.57	0.23	0.03	1.99	0.63	0.32	0.06

intermediate field ($\alpha_K = 2$) 2, and the evenly spread field ($\alpha_K = 3.5$) 1.4: a sweep of α_K at fixed scale varies mean capability along with inequality, and a higher mean makes resources more binding on its own. Table S6 compares that sweep with one in which the scale is set to $2(\alpha_K - 1)/\alpha_K$, so that every field has mean capability 2 (the realized means in the table fall short of 2 in the heavy-tailed field because the distribution has infinite variance; SI Materials and Methods). Holding the mean fixed narrows the heavy-tailed field’s advantage but preserves the ordering at every noise level; Fig. 2C and every number in the main text use the fixed-mean sweep.

Review compared at equal informativeness. The same review noise τ_K is a more informative signal where capability is more dispersed, because the noise then moves fewer researchers past one another. Table S7 therefore compares fields at equal informativeness, measured by the rank correlation between the review score

TABLE S7. The value of review by field at equal informativeness. Share of the no-review shortfall recovered when the rank correlation between the review score and capability is held at the stated value; fixed mean capability; interpolated within the noise sweep of Table S6.

α_K	rank correlation	0.8	0.55	0.3
1.3	0.94	0.80	0.50	
2	0.85	0.65	0.32	
3.5	0.66	0.46	0.18	

and capability, interpolating within the fixed-mean sweep. At a rank correlation of 0.8, 0.55, and 0.3, the heavy-tailed field recovers 0.94, 0.80, and 0.50 of the shortfall, the intermediate field 0.85, 0.65, and 0.32, and the evenly spread field 0.66, 0.46, and 0.18. The ordering is unchanged; the gaps between fields are smaller than at equal noise.

The association between capability and resources, the resource tail, and pool size. Table S8 varies the three remaining population parameters one at a time, at the default review noise and at Pareto scale 1. The association between capability and resources is the one that matters. As the copula parameter ρ rises from -0.5 to 0.8 , the value of targeting falls in every field (from 0.96 to 0.69 in the heavy-tailed field, from 0.25 to 0.06 in the evenly spread one), and so does the value of review; the ordering of fields is unchanged at every ρ . The mechanism is the gap rule itself: when resources already sit with the capable, the gaps $cK_i - R_{i0}$ are small and even, and there is little for targeting, or for the information that targeting requires, to add. The tail of the resource distribution changes the value of targeting by at most a tenth and the value of review by less. Pool size raises the value of targeting in the heavy-tailed field, because a larger pool contains more of the rare researchers of very high capability, and raises the value of review in the evenly spread field, where a larger pool gives review more to separate; neither effect changes the ordering.

The resource signal. The main text holds the resource signal at noise $\tau_R = 1$. Table S9 varies it from nearly exact ($\tau_R = 0.1$) to nearly uninformative ($\tau_R = 10$) and removes it altogether. Resource information matters least where review matters most. In the heavy-tailed field the no-review shortfall moves from 0.44 to 0.48 of the complete-information gain across the whole range, and the value of review from 0.79 to 0.76: the researchers who matter there are identified by capability, and resources add little to their identification. In the evenly spread field a nearly exact resource signal cuts the no-review shortfall from 0.15 to 0.09, because once resources are known, research output identifies capability; there the

TABLE S8. The value of targeting (upper block) and of review (lower block) by field as the copula parameter ρ coupling capability and resources, the tail of the resource distribution, and the size of the applicant pool vary. Review noise $\tau_K = 1$; Pareto scale 1; 50 simulated populations per entry; $M = 1,000$ particles. Default values: $\rho = 0$, $\alpha_R = 2$, $n = 50$.

α_K	copula parameter ρ				resource tail α_R			pool size n			
	-0.5	0	0.5	0.8	1.3	2	3.5	25	50	100	200
<i>Value of targeting (share of uniform funding's gain)</i>											
1.3	0.96	0.91	0.82	0.69	0.92	0.91	0.82	0.81	0.91	1.06	1.15
2	0.56	0.49	0.38	0.25	0.43	0.49	0.46	0.44	0.49	0.53	0.57
3.5	0.25	0.20	0.13	0.06	0.17	0.20	0.18	0.20	0.20	0.21	0.22
<i>Value of review (share of the no-review shortfall recovered)</i>											
1.3	0.92	0.90	0.88	0.85	0.88	0.90	0.92	0.87	0.90	0.94	0.95
2	0.67	0.65	0.62	0.56	0.65	0.65	0.65	0.60	0.65	0.71	0.74
3.5	0.24	0.22	0.16	0.01	0.24	0.22	0.21	0.14	0.22	0.27	0.32

TABLE S9. The resource signal. The no-review shortfall (upper block) and the value of review (lower block) as the noise of the resource signal τ_R varies from nearly exact to nearly uninformative, and with no resource signal; review noise $\tau_K = 1$; fixed mean capability; 50 simulated populations per entry; $M = 1,000$ particles.

	resource-signal noise τ_R					none
α_K	0.1	0.3	1	3	10	
<i>No-review shortfall (share of the complete-information gain)</i>						
1.3	0.45	0.44	0.45	0.47	0.48	0.48
2	0.25	0.25	0.27	0.29	0.30	0.30
3.5	0.09	0.09	0.11	0.14	0.15	0.15
<i>Value of review (share of the no-review shortfall recovered)</i>						
1.3	0.79	0.79	0.79	0.77	0.77	0.76
2	0.68	0.67	0.65	0.61	0.59	0.59
3.5	0.39	0.37	0.31	0.26	0.24	0.24

resource signal does a part of what review does, and the value of review falls from 0.39 with an exact resource signal to 0.24 with none. The ordering of fields is unchanged at every τ_R .

Multiplicative review noise. The main text's review noise is additive, so a given τ_K is a smaller relative error for a researcher of high capability than for one of low capability, and the question is whether that form favors the heavy-tailed field. Table S10 replaces the score by $K_i \exp(e_i)$ with e_i Gaussian of standard deviation s , a constant relative error, with the funder knowing the form. The form does matter, in the direction the question anticipates. At equal informativeness,

TABLE S10. Multiplicative review noise. The review score is $K_i \exp(e_i)$ with e_i Gaussian of standard deviation s , and the funder knows the form; fixed mean capability; 50 simulated populations per entry; $M = 1,000$ particles. The upper block gives the informativeness of the score at each s .

α_K	noise s				
	0.1	0.3	0.6	1	1.5
<i>Rank correlation between score and capability</i>					
1.3	0.96	0.83	0.65	0.48	0.35
2	0.93	0.73	0.51	0.34	0.24
3.5	0.85	0.55	0.33	0.21	0.14
<i>Value of review (share of the no-review shortfall recovered)</i>					
1.3	0.96	0.88	0.67	0.43	0.27
2	0.89	0.71	0.42	0.21	0.09
3.5	0.68	0.39	0.15	0.05	0.02

multiplicative noise costs the heavy-tailed field most: at a rank correlation of 0.65 between score and capability, review recovers 0.66 of the shortfall under multiplicative noise against 0.87 under additive noise in the heavy-tailed field, 0.61 against 0.75 in the intermediate field, and 0.49 against 0.55 in the evenly spread field, because a constant relative error is a large absolute error for exactly the researchers who matter most in a heavy-tailed field. The ordering of fields holds at every s and at every matched informativeness, and so does the main text’s statement that noisy review retains most of its value under heavy tails: at $s = 0.6$, review retains 69 percent of its maximum value in the heavy-tailed field, 47 percent in the intermediate field, and 22 percent in the evenly spread field. The heavy-tailed field’s advantage is smaller under multiplicative noise than the main text’s additive figures show, and it does not disappear.

Review observed once against review redrawn every round. In the main text each researcher’s review score is observed once and reused in every round, so a reviewer’s misjudgment persists (SI Materials and Methods). Table S11 redraws the score every round with independent noise, so that repeated review accumulates information the way the record does. Over two rounds the second review adds 0.01 to 0.04 to the share of the shortfall recovered, most where review is noisiest, and the ordering of fields is unchanged. Over twenty rounds the difference is larger: in the intermediate field the review funder’s shortfall falls from 8 percent of the complete-information gain in round one to 3 percent by round twenty when the score is observed once (Fig. 2B), and to 0.8 percent when it is redrawn every round; in the heavy-tailed field, to 2 percent and to 0.7 percent. The no-review funder’s shortfall in the same runs falls from 46 to 19 percent (default) and from

TABLE S11. Review observed once against review redrawn every round. Share of the no-review shortfall recovered over two rounds at review noise $\tau_K = 0.3, 1$, and 3; fixed mean capability; 100 (once) and 50 (redrawn) simulated populations per entry; $M = 1,000$ particles.

α_K	observed once (main text)			redrawn every round		
	$\tau_K = 0.3$	1	3	0.3	1	3
1.3	0.94	0.79	0.50	0.95	0.82	0.52
2	0.86	0.65	0.30	0.88	0.68	0.32
3.5	0.63	0.32	0.06	0.66	0.36	0.07

TABLE S12. The value of review when the review score partly reflects resources. Share of the no-review shortfall recovered when the score is $K_i + \beta R_{i0}$ plus noise, for $\beta = 0$ (the main text’s signal), 0.25, 0.5, and 1; fixed mean capability; 50 simulated populations per entry (100 at $\beta = 0$); $M = 1,000$ particles.

α_K	$\beta = 0$	$\tau_K = 0.3$				$\beta = 0$	$\tau_K = 1$		
		0.25	0.5	1			0.25	0.5	1
1.3	0.94	0.92	0.88	0.76	0.79	0.77	0.75	0.70	
2	0.86	0.82	0.75	0.55	0.65	0.62	0.59	0.52	
3.5	0.63	0.55	0.42	0.26	0.32	0.27	0.23	0.17	

60 to 14 percent (heavy-tailed) either way. Repeated independent review would therefore strengthen the main text’s comparison, and the once-observed score is the conservative case.

Review that partly reflects resources. The main text’s review signal observes capability with noise. Real review may also reflect the resources a researcher visibly has. Table S12 replaces the signal by $K_i + \beta R_{i0} + \xi_i$, with the funder knowing β and updating accordingly. Contamination costs least where capability is heavy-tailed: at $\beta = 1$, where the score weights resources as heavily as capability, the heavy-tailed field still recovers 0.70 to 0.76 of the shortfall, against 0.79 to 0.94 for a clean signal. It costs most where capability is evenly spread, where at $\beta = 1$ review recovers 0.17 to 0.26 of the shortfall, against 0.32 to 0.63 for a clean signal. The reason is the one behind Fig. 2C: the researchers whom review must find in a heavy-tailed field stand so far apart in capability that a resource term does not hide them.

The inequality of the gaps organizes the results. The runs above vary the population parameters in five ways. One quantity summarizes what they do to the value of targeting: how unequally the optimal grants, the gaps $\max(cK_i - R_{i0}, 0)$ at the population’s budget, are distributed across the field. Across all 159 population

settings in this appendix (mean capability, association, resource tail, pool size, and the two noise sweeps), the logarithm of the value of targeting is a nearly linear function of the Gini coefficient of the optimal grants, with $R^2 = 0.96$ (Fig. S7); adding the ratio of mean capability to mean resources raises R^2 to 0.99. Settings that differ in the capability tail, the association between capability and resources, the resource tail, and pool size, but agree on the inequality of the gaps, have the same value of targeting. The value of review depends on the same quantity together with the signal’s informativeness: the log-odds of the share recovered, regressed on the Gini coefficient of the optimal grants and the log-odds of the rank correlation between score and capability, has $R^2 = 0.90$, and adding the capability tail as a separate variable raises it by less than 0.01. Capability inequality is the main driver of gap inequality in every parameter sweep we considered, which is why the main text can speak of it directly; the association between capability and resources is the second driver, and it works through the same channel. Two cautions apply to using this quantity. The gaps depend on the budget, through c , as well as on the population, so the inequality of the gaps is a property of a field and a funder together; and it is an unobserved quantity, computed from capabilities and resources the funder does not observe, so it organizes the model rather than serving as a statistic a funder can estimate directly. The main text’s Discussion lists the observable quantities that stand in for it.

SI Text 6. STRATEGIC TIMING OF FUNDING

The funder holds a fixed budget for the whole horizon and may time its spending freely. The strongest argument for spending early concerns fields whose researchers have few resources: a researcher without resources produces nothing, so the funder observes nothing and no capability grows, and money held for later rounds produces neither evidence nor growth in the meantime. The argument fails. A funder that spends everything in the first round allocates before observing any research output, and knows exactly as little as a funder that spends everything in the last round; in a field with no resources at all, the two produce identical expected research output. What the argument does establish is a complementarity: early spending enables observation and growth, and late spending exploits what early spending revealed. That favors putting some money early. It does not favor putting most of it early.

In our simulations, under complementarity, observation comes first and money second. Wherever capabilities compound at more than a negligible rate (ϵ above about 0.03), the forward-looking funder spends less than an even share in the first round and spends the bulk once research output has revealed who is capable; scarce baseline resources raise the early share and weaken the effect, but do

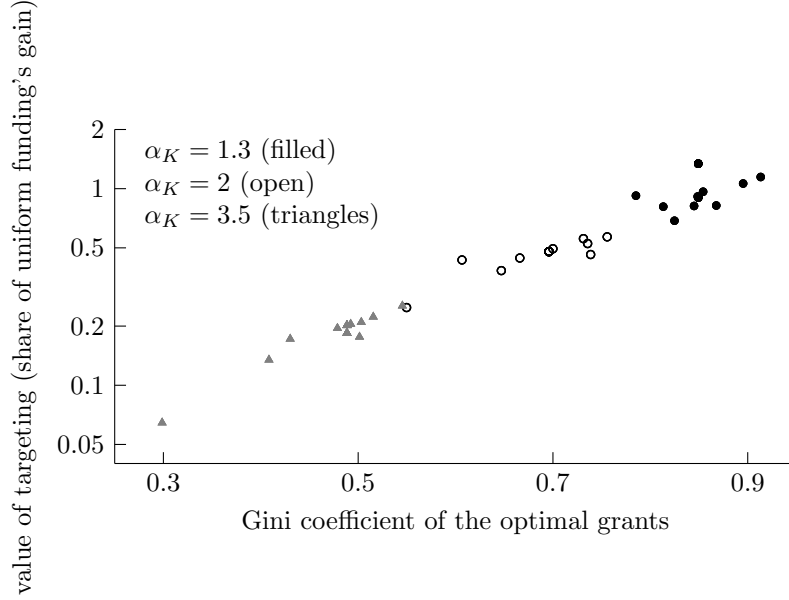


FIGURE S7. The inequality of the gaps organizes the value of targeting. Each point is one setting of SI Text 5: the value of targeting, on a logarithmic scale, against the Gini coefficient of the optimal grants at that setting's budget. The 159 settings vary the capability tail (marker), the correlation between capability and resources, the tail of the resource distribution, the size of the applicant pool, and the scale of the capability distribution; the logarithm of the value of targeting is linear in the Gini coefficient with $R^2 = 0.96$.

not reverse it (Fig. S8A). The shift of spending toward later rounds comes from the research output researchers produce whether or not they are funded, which compounds and reveals capability while the funder waits (SI Text 3, Fig. S4).

Planning ahead is worth less than the signal provided by grant peer review, in every setting we test. Across all horizons and compounding rates in our sweeps, the forward-looking funder's gain over the myopic funder, which spends equal installments each round, stays below the review signal's value: by an order of magnitude or more at the default compounding rate and below, and approaching the signal's value only at the highest compounding rate and the longest horizon we consider (Fig. S8B). As a share of the research output that the forward-looking funder's funding adds, the gain from planning ahead ranges from indistinguishable from zero to 4.5% across the 30 settings tested; the review signal's value in the same settings ranges from 7% to 27%. The gain exists only where capability and resources are complements: under Cobb-Douglas production it is indistinguishable from zero (SI Text 4, Table S5).

Three limits bound these results, and they are the reason the timing analysis sits in the SI rather than the main text. First, the forward-looking funder is a certainty-equivalent planner (SI Materials and Methods): it treats future research output as equal to its expectation and does not value the information its own grants will generate, so it is not the Bayes-optimal sequential policy, and its gain is a lower bound on what planning ahead could be worth. Second, the gain combines two changes at once, when the money is spent and whom it reaches, because a funder that anticipates capability growth also values recipients differently; we do not separate the two, and “gain from planning ahead” names their sum. Third, the objective counts research output within the horizon only and assigns no value to the capability that remains at its end. A terminal value for capability would credit the capability that late grants build, which the horizon objective ignores, and also the compounded capability that early grants leave behind; we have not measured which effect is larger, so the size of the shift toward later rounds should be read as specific to the horizon objective.

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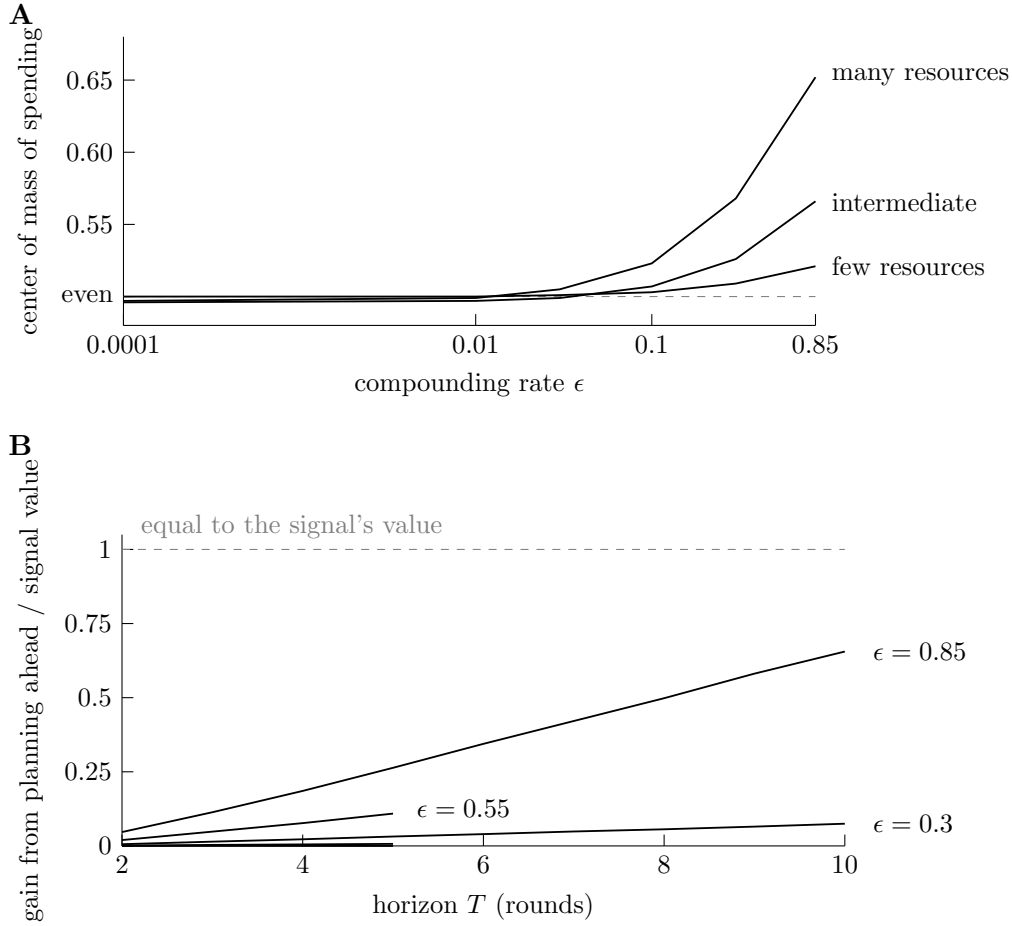


FIGURE S8. Strategic timing of funding. (A) The compounding rate, not the scarcity of baseline resources, sets how far spending shifts toward later rounds: the center of mass of spending (0.5 is even; higher is later) as the compounding rate ϵ increases, for three baseline resource scales. (B) Choosing whom outweighs choosing when: the gain from planning ahead, as a share of the review signal's value, by horizon and compounding rate; in every setting the gain is below the signal's value. Settings: SI Materials and Methods.

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